46 CHAPTER Lumbar Disc Disease

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Introduction

The location of the anatomic pain generator in patients with low back pain is often difficult to discern. Pain can originate from several anatomic structures within the spine, making it difficult for both the patient and physician to localize. The intervertebral disc (IVD) is a widely accepted source of chronic low back pain (LBP). A study reviewing magnetic resonance images (MRIs) of patients with persistent LBP showed disc degeneration in 87% of participants.[1] Additionally, patients with severely degenerate discs are 3.2 times more likely to have LBP.[2] Despite this robust link between disc degeneration and back pain, degeneration is a normal consequence of aging that does not routinely cause discomfort, even with a corresponding reduction in biomechanical function.[3,4] However, there is still much work to be done to identify the underlying pathology that distinguishes painful degenerate disc disease from nonpainful age-related changes.

Common subgroups of LBP of IVD origin include lumbar disc herniation, internal disc disruption, and degenerative disc disease. Lumbar disc herniation causes an acute onset of pain, while degenerative disc disease is linked to chronic LBP. Degenerative disc disease also contributes to the pathogenesis of secondary spinal disorders such as spinal stenosis and degenerative spondylolisthesis. Improved understanding of the pathophysiology of IVD disorders has led to a resurgence of enthusiasm in the development of new pharmacobiologics and treatment techniques for this common disorder. New advances in physical therapy and operative technologies, such as IVD replacement and minimally invasive surgical techniques, are challenging traditional methods of treatment.

This chapter discusses the anatomy, pathophysiology, diagnosis, and treatment of primary disc disorders, namely internal disc disruption (IDD) and degenerative disc disease (DDD). Disorders arising secondarily from the IVD are beyond the scope of this chapter.

Natural History

Sixty to eighty percent of the adult population can be expected to experience LBP at some point during their lifetime. The annual incidence of back pain in the adult population is 15%, and its point prevalence is about 30%.[5] By the age of 30 years almost half the population will have experienced a substantive episode of LBP.[6] Fortunately, the vast majority of symptoms are short lived; it is generally believed that 80% to 90% of patients with episodes of LBP will recover within 6 weeks of onset regardless of the type of treatment.[7]

Even though resolution of symptoms is the common and expected outcome, there is also a high recurrence rate. Croft et al.[8] reported that although 90% of subjects stopped pursuing treatment for their symptoms within 3 months, most still had substantial LBP and related disability. In addition, only 25% of the patients who sought consultation for LBP had fully recovered within12 months. In a survey of the British general population, 38% of adults reported a significant episode of LBP within a 1-year period, of which one-third had experienced symptoms for longer than 4 weeks.[9] Inability to return to work within 3 months of symptom onset is a poor prognostic indicator. Only 20% of patients still disabled after 1 year will return to work, and only 2% return after 2 years.[10]

The clinical onset and course of LBP may be prolonged for many patients and may best be represented as a continuum of back-related disability and distress.[11] A significant number of patients presenting with acute LBP have a prior history of chronic back pain.[12] The strongest predictive factor for a new episode of LBP is a previous episode.[13,14]

The natural history of DDD is largely unknown. Smith et al.[15] reported on the outcome of 25 discogram-positive patients treated nonoperatively and found that 68% of patients improved by the 3-year minimum follow-up. Although 60% of the patients were involved with worker's compensation and 32% were being treated for psychiatric diagnoses, this study suggests that at least two-thirds of those with discogenic pain improve with conservative therapy. The retrospective study design and small sample size limit the conclusions, and since only patients with significant symptoms typically undergo discography, the natural history of untreated, less-severe cases of symptomatic DDD is likely to result in more improvement than the cited study.

In their classic description, Kirkaldy-Willis et al.[16] classified the degenerative process into three distinct phases: dysfunction, instability, and stabilization. In the first phase the disc loses its normal function as the degenerative process begins. This is followed by a period of relative instability as degeneration progresses with intermittent episodes of pain. During the instability phase abnormal motion can occasionally be seen on flexion-extension radiographs; however, the spinal segments during this phase of degeneration more often show no demonstrable radiographic instability. The final phase—stabilization—results when the spinal segment has reached a new equilibrium because of loss of height and compression of disc tissue; at this point the patient typically no longer has episodes of back pain. A problem with this theory is that patients who meet either the radiographic diagnostic criteria of DDD—loss of height, osteophytes, or even olisthesis—or show the MRI signs of disc degeneration—signal changes when compared with adjacent levels—can be completely symptom free. Waris et al.,[17] in a study with 17 years of follow-up MRIs, showed that young patients with DDD did show radiographic evidence of progression, but it was not significantly associated with LBP or a higher rate of surgery.

Relevant Anatomy

The major functional role of the IVD is mechanical; it supports the compressive load of the torso while allowing for polyaxial flexion, extension, and torsion. Although the movement capable of a single spinal motion segment is quite limited, the range of motion supported by the 24 presacral motion segments of the spine in its entirety is remarkable. Found between two rigid vertebral bodies, the disc allows this movement while simultaneously supporting compressive loads. The discs themselves are complex organs comprising central proteoglycan-rich nucleus pulposus surrounded circumferentially by fibrocartilaginous anulus fibrosus. The nucleus pulposus is bound caudally and cephalically by the cartilage end plates of the contiguous vertebrae. Major characteristics of these tissue components of the disc are highlighted below.

Nucleus Pulposus

The central nucleus pulposus is derived from the embryonic notochord, and notochordal cells remain in the tissue after birth and into adult life. During development, the nucleus is highly cellular; after birth and an initial period of cell growth, the number of cells declines. In the adult, the cell density is very low.[18] The histology of the nucleus pulposus cells is unique and complex; large cells are arranged mainly in clusters separated by an abundant extracellular matrix. Among the large notochordal cells, much smaller cells, possibly derived from the notochordal sheath, can also be seen.[19] The large cells appear to have numerous vacuoles, which have prompted some to describe them as "physaliphorus."[20] Transition electron microscopic analysis showed that the nucleus pulposus contained cell clusters embedded in a proteoglycan-collagen matrix. The cells exhibit a well-defined Golgi system, an extensive endoplasmic reticulum, and a complex vesicular system filled with beaded structures. Neither necrotic nor apoptotic cells are evident. Although a defining characteristic of the cells is the presence of numerous cytoplasmic processes, a remarkable finding is that the cells contain few, if any, mitochondria. Moreover, since the nucleus pulposus has no blood supply, the oxygen tension within the disc is very low. These limitations have prompted one study group to note that nucleus pulposus cells "tune" their metabolism to the available oxygen supply.[21] In this case, nucleus pulposus cells rely to a large extent on the glycolytic pathway to generate metabolic energy.[22]

With respect to the extracellular matrix, nucleus pulposus cells secrete aggrecan as well as collagen II and, to a very small extent, collagen I. The matrix also contains collagen IX and XI, and collagen X has also been reported to be present during degeneration. Because of the presence of hydrophilic aggrecan, the disc exhibits a high osmotic pressure. The high water content of a healthy nucleus pulposus, as evidenced by the intense bright signal on T2-weighted MRI, is responsible for supporting compressive loads.

Anulus Fibrosus

Compressive deformation of the nucleus pulposus is limited circumferentially by the anulus fibrosus, which experiences the compressive loads of the torso as hoop stress. Since the anulus fibrosus experiences tensile stress as opposed to compression, its anatomy and biochemical composition are distinct from the nucleus pulposus. The anulus is not a homogeneous tissue; it is divided into an inner region and an outer, or peripheral, fibrous zone.[23] The outer anulus fibrosus is composed of very well-defined collagen I fibers that bundle to form long parallel concentric lamellae. Marchand and Ahmed[24] showed that the number of fiber bundles varies from 20 to 62. The thickness of lamellae varies both circumferentially and radially. Age, location, and vertebral type also influence lamellar architecture. The central anulus fibers insert into the end plate cartilage, whereas those at the periphery anchor to the vertebral bone.

The inner anulus fibrosus represents approximately 50% of the total radial thickness. Many consider the inner anulus a transition zone because it differs substantially from the outer region. Compared with the outer anulus fibrosus, where the cells are elongated and fusiform aligned with the long axis of the fibrils, the cells of the inner anulus are spherical in shape and more closely resemble chondrocytes. These cells are few in number with short processes. A further difference between the inner and outer anulus is their chemical composition. The inner anulus contains collagen I and II as well as versican. Although aggrecan is present in both regions of the anulus, decorin and biglycan are found mainly in the outer anulus. The other protein of significance is elastin, which accounts for 2% of the dry tissue weight and aids the anulus in its role of withstanding cyclic tensile forces.

End Plate Cartilage

The third component of the disc is end plate cartilage, which is located at the junction of the vertebral body and the disc. This thin layer of hyaline-like cartilage is most thick in the newborn and thins with age. It serves as an interface between the soft nucleus pulposus and the dense bone of the vertebrae as well as a biomechanical barrier that prevents the disc from applying pressure directly to the bone. In addition, the end plate cartilage plays a role in maintaining nucleus pulposus cell viability and controlling biosynthetic activities.[18,25] In the adult human, the width of the end plate is up to 1 mm. It contains chondrocytes embedded in an aggrecan-rich and collagen II extracellular matrix. Although the cells do not undergo terminal differentiation, collagen X may be present in the central region of the end plate, perhaps in relationship to focal areas of endochondral bone formation. The end plate transitions into bone through a region of calcified cartilage. Work in rabbit and rat models suggests that end plate chondrocytes may contribute a cell fraction that makes up some of the nucleus pulposus and the anulus fibrosus in normal development and during injury.[26,27] However, cell lineage studies are required before a definitive contribution of such a homeostatic mechanism is understood. The presence of the cartilage layer provides the motion segment with its jointlike characteristics.

In his review on the end plate, Moore[28] noted that vascular channels penetrate the cartilage during development, but at maturity the vessels become narrow or even obliterated. It is likely that this change impacts the nutrient supply to both the cartilage and the disc. Nachemson et al.[29] showed that, at the tissue periphery, the cartilage is much less permeable to low-molecular-weight dyes. Supporting this, Crock and Yoshizawa[30] showed that the central region of the end plate, where there is a high concentration of channels, is freely permeable to small molecules. Clinically, it is not uncommon to note that the central region undergoes sclerosis or mineralization with alterations in the mechanical properties of the cartilage. When this occurs, nucleus pulposus tissue can be forced through the end plate into the underlying bone of the vertebrae. This phenomenon is referred to as Schmorl's nodes, which Schmorl himself considered to be linked to degenerative changes at the cartilage-bone interface.

Defining the IVD Niche: Molecular Characters

One overriding aspect of disc cell biology is that nucleus pulposus and cells residing in the inner anulus are removed from the blood supply. Blood vessels originating in the vertebral body traverse the superficial region of the end plates, but none of these vessels infiltrates the nucleus pulposus. Gruber and colleagues[31] showed that the anulus is avascular except for small capillary beds in the dorsal and ventral surfaces, and this vasculature never enters the nucleus pulposus.[32,33] The lack of penetrating vasculature suggests that the disc is a low-oxygen environment. Studies have confirmed that this logic holds true in vivo. Experimental measurements done in canine discs[34] as well as modeling studies by Bartels et al.[35] indicate that the pO_2 within the disc is indeed low. Thus, it is now widely recognized that the nucleus pulposus cells reside in a hypoxic niche.[36]

However, this raises an important question: How do cells survive in this unique niche? Due to pioneering studies by Semenza and others, it is now evident that the key molecule regulating energy metabolism and survival activity under hypoxia is hypoxia-inducible factor 1 (HIF-1),[37] a member of the basic helix-loop-helix (bHLH)–PER-ARNT-SIM family of proteins. HIF-1 is composed of a constitutively expressed β-subunit and a regulatory α-subunit. The α-subunit is stable under hypoxia but is rapidly degraded in normoxia.[38] Transactivation of HIF-1 target genes involves dimerization of the two subunits and binding to an enhancer, the hypoxia-response element in target genes. HIF-1 serves as a key transcription factor that regulates the expression of glycolytic enzymes, and enzymes that control the activity of the tricarboxylic acid cycle and oxidative phosphorylation.[37,39,40] Additional target genes include those that contribute to apoptosis, autophagy, and matrix synthesis pathways.[41-43] Details of these relationships are shown schematically in Fig. 46.1. Other isoforms of HIF exist, the most important being HIF-2α. Recent evidence suggests that HIF-1α and HIF-2α are not redundant, and that the relative importance of each of the homologues, in response to hypoxia, varies among different cell types.[44] In addition to these genes, the Sox family of transcription factors that are essential for the development and function of the nucleus pulposus are hypoxia and HIF sensitive.[45-48] However, the relationship between Sox proteins and HIF in the hypoxic niche of the IVD is not yet clearly demonstrated. The influence of HIF on cell function is vast, warranting further

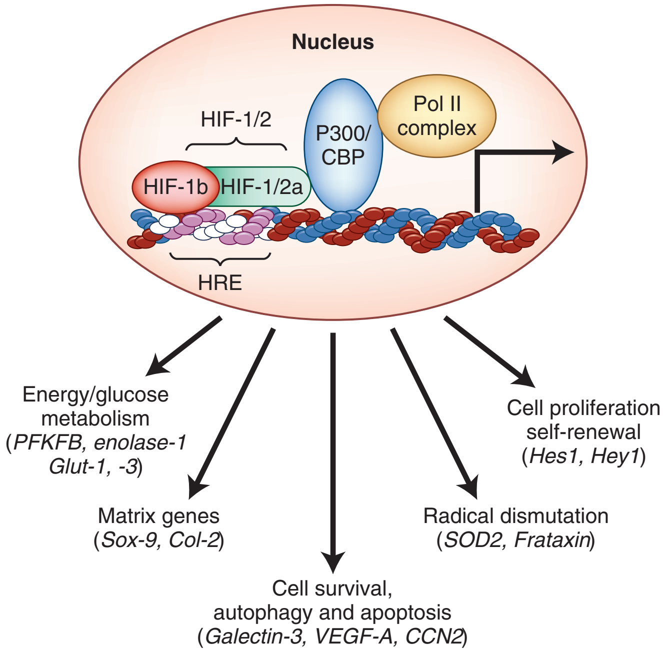

FIG. 46.1 Functional activity of hypoxia-inducible factor (HIF) target genes. Critical functions include energy metabolism, angiogenesis, cell survival, autophagy and apoptosis, matrix synthesis, proliferation, self-renewal and differentiation, radical dismutation, and pH regulation. Many of these functions are critical for survival and functioning of the nucleus pulposus cells in the avascular niche of the intervertebral disk. Hypoxia/HIF-sensitive proteins that are identified in the nucleus pulposus cells are shown in parentheses. (Modified from Risbud MV, Schipani E, Shapiro IM, et al. Hypoxic regulation of nucleus pulposus cell survival: from niche to notch. *Am J Pathol.* 2010;176[4]:1577–1583.)

study, but it is evident that this protein is responsible for nucleus pulposus survival in the hypoxic disc.

A number of reports showed that there is a robust HIF response by nucleus pulposus cells. The response is evident across species; more importantly, HIF-1α activity is unresponsive to the oxemic state of the tissue.[22,36,49] Accordingly, when compared with most other tissues, there are important underlying differences in the HIF status and reactivity of disc cells. HIF-1α expression and activity is always "on." This unique response suggests that stabilization of HIF-1α in nucleus pulposus cells ensures that transcriptional activity is a major governor of their function. The second HIF homologue, HIF-2α, is robustly expressed by nucleus pulposus cells. Like HIF-1α, steady-state protein levels are similar in both hypoxia and normoxia, suggesting that it is stabilized as well.[50] Recent work by Merceron and colleagues[51] has unequivocally demonstrated that HIF-1 is indispensable for nucleus pulposus cell survival in vivo. Conditional deletion of HIF-1α in mouse notochord by FoxA2-driven Cre recombinase results in massive apoptotic cell death in the nucleus pulposus at birth, likely due to energetic failure by decreased glycolysis. Eventually the nucleus pulposus is replaced with a fibrocartilaginous tissue with inferior biomechanical properties. Interestingly, lineage-tracing studies clearly showed the notochordal lineage of the nucleus pulposus cells, and their replacement with nonnotochordal cells, likely migrate into the tissue from the inner anulus or end plate cartilage (Fig. 46.2). Thus, because of the necessity of HIF-1α in maintaining nucleus pulposus cell survival and function, the constitutive expression of HIF-1α is considered a marker of their phenotype.[52] Before leaving this topic, it is important to comment that a growing body of literature now points to a unique relationship between HIF-1α and the O_2-sensing prolyl hydroxylase, enzymes that control HIF-α stability and activity. HIF is a key player in nucleus pulposus cell function and survival. Consequently it is likely that disturbance in the HIF–prolyl hydroxylase circuit would compromise cell function and exacerbate disease.

Another important niche condition that defines the IVD is the high proteoglycan content accounting for the elevated water content in these tissues; the percentage water of the nucleus pulposus and anulus fibrosus is approximately 77% and 70%, respectively.[53] Several reports indicate that the IVD is hyperosmolar when compared with other tissues. Values reported vary from 430 to 496 mOsm.[54,55] This unusually high extracellular osmotic pressure affects both cell function and matrix synthesis. Ishihara and colleagues were the first to demonstrate this relationship in disc tissue using bovine nucleus pulposus explants.[54] How extracellular osmolarity regulates the expression levels of specific matrix molecules has received intense study in recent years. In human nucleus pulposus and anulus cells, expression levels of aggrecan and collagen II were increased in cells under hyperosmotic conditions (500 mOsm), whereas collagen I expression was downregulated.[56] Similarly, in bovine nucleus pulposus cells, an increase in medium osmolarity from 300 to 500 mOsm increased aggrecan expression and decreased levels of matrix metalloproteinase (MMP)-3 mRNA.[57] To relate osmolarity to

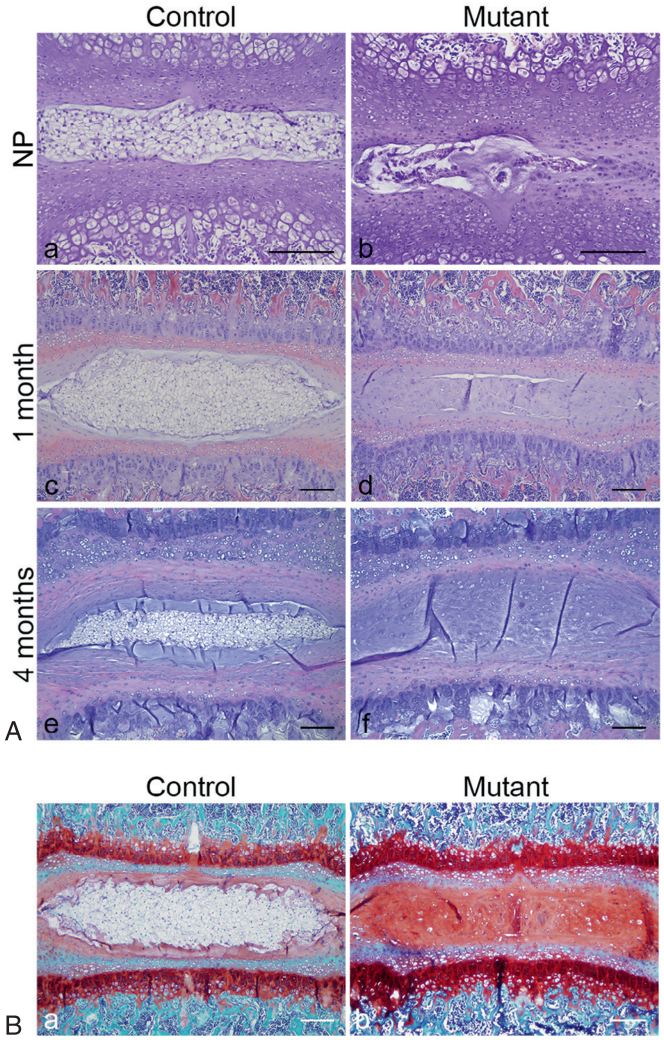

FIG. 46.2 Progressive disappearance of the nucleus pulposus (NP) postnatally in mice with conditional deletion of HIF-1α in NP. (A) Hematoxylin and eosin staining of newborn *(a,b)* and at 1 month *(c,d)* and 4 months *(e,f)*. Results are shown for control (Foxa2^{Cre};HIF-1α$^{f/+}$) and mutant (Foxa2^{Cre};HIF-1α$^{f/f}$) mice. Bar = 50 μm. (B) Safranin O staining of NP at 1 month *(a,b)* in control (HIF-1α$^{f/f}$) and mutant (Foxa2^{Cre};HIF-1α$^{f/f}$) mice. Bar = 100 μm. (Modified from Merceron C, Mangiavini L, Robling A, et al. Loss of HIF-1α in the notochord results in cell death and complete disappearance of the nucleus pulposus. *PLoS One.* 2014;9:e110768.)

loading, rabbit discs were cultured under hyperosmotic conditions (485 mOsm 8 h/day); although loading did not influence proteoglycan content or disc cell viability, after 28 days in culture the raised osmotic pressure prevented aberrant overexpression of collagen I and appeared to influence aggrecan expression.[58]

Loading of the hydrostatically pressurized IVD results in minute-to-minute fluctuations in extracellular osmolarity.[59,60] From this perspective, it is not surprising that tonicity enhancer binding protein (TonEBP), the only known mammalian transcription factor responsive to changes in osmolarity, is robustly expressed in both nucleus pulposus and anulus fibrosus tissues.[61] Importantly, decreased expression of TonEBP correlated to increased evidence of cell death, suggesting that this protein is necessary for nucleus pulposus cell viability in the hypertonic milieu of the disc. Later studies showed that TonEBP controls expression of critical matrix-related targets as well as the water channel AQP2.[62,63] Analogous to the importance of HIF for the survival of the disc due to its hypoxic niche, TonEBP may prove to play a central role in disc survival in the hypertonic stress. For detailed discussion of osmosensing in intervertebral disc see recent review by Johnson et al.[64]

In addition to the above niche conditions, the IVD is also largely aneural, with innervation in a healthy disc limited to the peripheral fibers of the anulus. The sinuvertebral nerve innervates the disc, posterolateral ligament, ventral dura, posterior anulus, and blood vessels. It is composed of a sensory branch from the ventral root and a sympathetic branch from the gray rami communicans near the distal pole of the dorsal root ganglion. The sinuvertebral nerve is believed to have three segmental levels of overlap, which makes it difficult to localize pain originating in the disc, dura, and posterolateral ligament. Nakamura et al.[65] treated 33 patients with a selective L2 nerve root block and showed good relief of back pain. These authors hypothesized that the main afferent pathways of pain from lower lumbar IVDs in patients with discogenic back pain are sympathetic in nature and are mediated through the L2 nerve root via the sinuvertebral nerve; this hypothesis has yet to be validated, however.

Changes in Disc Structure With Aging and Degeneration

Changes that almost universally occur in the IVD with aging include reduction in disc volume, shape, and content. Alterations in gene expression and transcription factors may be responsible for cell senescence within the disc. These senescent cells lose biochemical and synthetic capabilities, which ultimately diminishes the ability of the disc to recover from deformation and renders the matrix more vulnerable to progressive fatigue failure. The nucleus pulposus gradually becomes less hydrated, and by the third decade of life there is usually already a significant decline in the number of viable cells and a loss of proteoglycans.

The early degenerative process affects the nucleus pulposus and the end plate more than the anulus fibrosus. Both anabolic and catabolic processes are upregulated during the early stages of degeneration; however, anabolic repair processes fail to keep up with the catabolic processes and matrix degeneration ensues over time. As the process progresses, the inner layers of the anulus and the nucleus pulposus gradually become indistinguishable and change into a stiff desiccated fibrocartilaginous material.

The number of arterioles supplying the peripheral disc diminishes significantly as remaining blood vessels are obliterated by calcification of the cartilaginous end plates. Loss of end plate vascularity and porosity leads to a reduction in the influx of nutrients and efflux of waste products. Lactate levels rise locally within the hypovascular disc secondary to increased production and decreased removal. Cell apoptosis occurs as a result of decreased tissue pH,[66] and the biosynthetic reparative capability of the disc is thus further impaired.

Thinning and microfracture of the end plate alter its diffusion properties. This may allow rapid outflow of fluid through the cartilage end plate upon loading, thereby rendering the hydrostatic pressure mechanisms involved in load transference less effective and uniform. Focal elevations in shear stresses at the disc level may further adversely affect the disc structure and result in anular damage. Over time cracks develop between and through the anular lamellae. Eventually communicating channels may develop between the peripheral layers of the anulus and the nucleus, and disc material can then herniate through these fissures. The weakened anulus can develop a full-thickness defect and allow near-complete herniation of the nucleus pulposus, particularly when the disc is loaded in flexion and torsion.

The degenerative process resulting from matrix changes and internal structural disruption sets the stage for abnormal motion at the degenerated segment. Changes in disc structure alter the loading response and alignment of the spinal column. These changes can influence the facet joints, ligaments, and paraspinal muscles, which may also become pain generators. However, pain does not always correlate with morphologic changes in the disc and mechanical compression.[67] MacNab[68] described traction osteophytes around the vertebrae originating 2 mm from the anterior end plate at the site of attachment of the outermost anular fibers. These osteophytes were thought to be signs of abnormal biomechanics caused by traction at the insertion of the anular fibers into the vertebral bodies. Subsequent studies found these osteophytes to be inconsistently present.

Thus multiple factors lead to disc degeneration, including insufficient nutritional supply, reduction in the amount of viable cells, degradative enzymatic activity, and cell senescence and apoptosis. Alteration in loading patterns between the end plate and disc leads to anular damage and the potential for herniation. Perturbation of the disc also leads to degeneration and pain in other segmental structures such as the facet joints, ligaments, and paraspinal muscles. The initiating event(s) leading to the onset of degeneration remains unknown.

Associated Factors

Various risk factors have been implicated in the pathogenesis of lumbar disc degeneration. Hangai et al.,[69] in a recent review of factors associated with IVD degeneration in the elderly, cited increased age, high body mass index, occupational lifting, sporting activities, and factors associated with atherosclerosis as risk factors. Multiple studies have demonstrated genetic contributions to degenerative LBP.[70] Battié et al.[71] estimated the familial contribution to IVD degeneration to be between 34% and 61%. Cigarette smoking has also been implicated and appears to have an adverse vasoconstrictive and atherosclerotic effect on the nutrition of the IVD.[72,73] Interestingly. effects of cigarette smoking on disc degeneration seem to be gender specific, with males, but not females, being prone to smoking-induced pathology.[74,75] Type of occupation has also been demonstrated to have an adverse effect on lumbar spinal segment degeneration, increasing the risk of symptomatic DDD. Studies have implicated occupations that require repetitive lifting or pulling, prolonged sitting[76] (such as motor vehicle driving[77]), and whole-body vibration.[78]

Arun et al.[79] used serial postcontrast MRI to study the effect of prolonged mechanical load on diffusion into the IVD. These authors reported that 4.5 hours at a load corresponding to 50% body weight significantly retarded the diffusion of small solutes into the center of the IVD, and 3 hours in an unloaded recovery phase were required to return the diffusion rate to that seen in the unloaded disc. Prolonged mechanical load can therefore cause a disruption of diffusion that may accelerate disc degeneration; however, this hypothesis is yet to be confirmed clinically.

The genetic predisposition to lumbar DDD and lifetime exposures were studied in a classic monozygotic twin study by Battié et al.[71] They reviewed 115 male identical twin pairs for exposures to common risk factors such as occupation, recreational activities, driving, and smoking. Disc degeneration was determined by MRI and clinical evaluation. In the upper lumbar spine only 7% of the variability was explained by occupation, 16% by age, and 77% by familial aggregation. In the lower lumbar spine recreational physical loading explained 2% of variability, age explained 9%, and familial aggregation explained 43%. The authors concluded that primarily genetic and other unexplained factors result in DDD, whereas commonly implicated environmental factors have only modest effects. In a 5-year follow-up study of the same twin population, the authors reaffirmed that genetics have a dominant role in the progression of DDD, whereas occupational lifting and leisure activity had only modest effects.[80] The important role of genetic factors has been corroborated in other twin studies[81,82] but appears to be less of an explaining factor for back pain in older populations.[83]

Several gene loci have been determined to be associated with increased risk for DDD. Type IX collagen was one of the first gene loci identified with some aberrant alleles imparting a 3- or 4-fold increase in relative risk.[84-86] More recent publications also implicate collagen type XI, interleukin-1, aggrecan, the vitamin D receptor, MMP-3, and corticotropin-like immediate lobe peptide as candidate genes.[87] In a meta-analysis of 4600 subjects using disc space narrowing and osteophyte growth as a continuous trait, Williams et al.[88] identified four signal nucleotide polymorphisms with a P value $< 5 \times 10^{-8}$. Among these identified signal nucleotide polymorphisms was a polymorphism in the intron of the Parkinson protein 2, E3 ubiquitin protein ligase (PARK2) gene on chromosome 6 (rs926849) significantly associated with lumbar disc degeneration, a subset of the patients also exhibited differential methylation at one CpG island in the PARK2 promoter, providing additional evidence indicating the importance of this gene to DDD. Another study looking at 4043 DDD patients and 28,599 controls identified carbohydrate sulfotransferase 3 (CHST3) as a susceptibility gene for disc degeneration based on enhanced microRNA binding. CHST3 catalyzes proteoglycan sulfation.[89] The discovery of these genetic risk factors has yet to inform new useful diagnostic and treatment modalities, but they offer novel unbiased insight into the pathophysiology of disc degeneration.

Pathophysiology

Internal Disc Disruption

Henry Crock coined the term "internal disc disruption" in 1970 and defined it as a painful increase in biologic activity of the IVD after injury with normal radiographic, computerized tomographic, and myelographic examinations but an abnormal discogram.[67] IDD as a cause of discogenic back pain remains controversial. The advent of MRI has dramatically improved the suspect presence of this entity; IDD presents as a dark disc with relatively preserved height and contour. Pain in IDD is believed to be caused by mechanical and chemical stimulation of nociceptors within the anulus or on the surface layers of the anulus and the overlying ligamentous tissue. The hallmark of IDD is the absence of disc herniation, prolapsed disc material, segmental instability, or other radiographic abnormality.[67,90] Nerve root irritation, radicular pain, and neurologic deficits are also absent. Radiographic changes associated with DDD—such as significant disc space narrowing, end plate osteophyte formation, end plate sclerosis, and gas formation within the disc space—are not seen in IDD.[91] MRI (dark disc), and positive discography (concordant pain in the abnormal level, not at normal adjacent levels) are required to make the diagnosis of IDD. Because of the poor sensitivity and specificity of discography, there are many disbelievers of the existence of IDD as a clinical entity. It is important to note that there is a growing consensus spearheaded by Carragee that discography may accelerate disc degeneration, loss of disc height, and overall progression of pathology and symptoms.[92] Advanced imaging technologies offer a noninvasive way to measure degeneration in intact discs. T1ρ MRI imaging can discriminate painful from nonpainful discs, and T1ρ values strongly correlate with in vivo opening pressure measurements obtained by discography.[93]

Degenerative Disc Disease

The relationship between DDD and LBP is not well understood. Two potential sources that have been implicated as contributors to discogenic pain are sensitization of nerve endings by release of chemical mediators and neurovascular ingrowth into the degenerated disc. The precise pathophysiologic mechanism for chemically mediated induction of hyperalgesia within the disc has yet to be fully elucidated. Radial anular tears provide a route for nuclear material and noxious chemicals to leak from the disc and contact the dural sac and nerve roots; some studies have shown that autologous nucleus pulposus alone has the capacity to produce a robust inflammatory response. The presence of nerve endings in the disc is a controversial topic; abundance of neurovascular ingrowth into the degenerate disc has been less substantiated by recent studies.[94] Some of the important features of the degenerating disc are described below.

Defining Features of the Degenerating Disc

Both nucleus pulposus and anulus fibrous cells are able to secrete a variety of inflammatory cytokines in response to various noxious stimuli and altered mechanics. During the degenerative process, disc cells secrete elevated levels of tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β), IL-6, and IL-17. These cytokines stimulate the production of additional cytokines, induce matrix catabolism, alter cell protein synthesis, and affect the biophysical properties of nucleus pulposus cells. Matrix destruction is mediated by cytokine-induced expression of MMP and a disintegrin and metalloproteinase with thrombospondin motifs (ADAMTS).[95] This system amounts to a positive feedback loop in which disc degeneration stimulates cytokine production, which in turn stimulates more disc degeneration and cytokine production.

IL-1β and TNF-α are the two most widely studied cytokines in the pathogenesis of disc degeneration. IL-1β is synthesized in an inactive pro-IL-1β form and activated by caspase-1 before secretion. IL-1β binds to IL-1β receptor that acts through the myeloid differentiation primary response gene 88 (MYD88) to induce expression of matrix-degrading enzymes.[52,96] Expression of both IL-1β and its receptor increases in correspondence to the severity of disc disease. Not only does IL-1β signaling induce matrix catabolism by inducing matrix-degrading enzymes, signaling also interferes with aggrecan and collagen II synthesis.[97] TNF-α, on the other hand, is synthesized as a type II transmembrane cytokine. TNF-α-converting enzyme cleaves the membrane-bound portion to generate secreted TNF-α. Nuclear factor-κB light chain enhancer of activated B cells and mitogen-activated protein kinase pathways are the two primary downstream targets of TNF-α signaling in disc cells (see Fig. 46.1).[52] Importantly, in addition to transcriptional induction of several catabolic mediators, TNF-α promotes ADAMTS-5 processing and activation by elevating cell surface levels of syndecan-4, a heparan sulfate proteoglycan.[98] A recent study has also shown the critical contribution of syndecan-4 in controlling TNF-α–mediated transcriptional induction of MMP-3.[99] In addition to their well-described catabolic functions, higher levels of inflammatory cytokines TNF-α and IL-6 have been shown to cause cell death in dorsal root ganglion neurons.[100] Moreover, IL-6 and IL-8 are correlated with painful degenerate discs.[101] Both IL-1β and TNF-α also upregulate nerve growth factor, with IL-1β also inducing the expression of brain-derived neurotrophic factor and substance P.[102-104] Thus, not only could these neurotrophic factors cause pain by dorsal root ganglion sensitization and through retrograde signaling, but substance P has been shown to upregulate synthesis of inflammatory cytokines (IL-6, IL-8, and TNF-α), further exacerbating to progression of disc degeneration.[105] Although it is evident that inflammatory phenotype characterizes degenerate discs, little is known about how the inflammation is initiated and sustained to give rise to chronic LBP.

Subclinical Bacterial Infection of the Spinal Motion Segment as a Possible Initiator of Low Back Pain

There is growing interest and controversy in the notion that subclinical anaerobic bacterial infection could play a role in symptomatic disc degeneration. This subclinical infection is in contrast to LBP from overt discitis, in which 80% of patients present with elevations in erythrocyte sedimentation rate and 50% present with a fever.[106,107] Changes in end plate radiologic appearance may lead to clues to identifying patients who have bacteria-mediated LBP. Modic et al.[108] used MRI to detect changes in vertebral bone marrow associated with DDD. Type 1 changes were described as decreased signal intensity on T1-weighted images with increased signal intensity on T2-weighted images; type 2 changes were described as increased signal intensity on T1-weighted images with slightly increased intensity on T2-weighted images. Fat appears bright and increases the signal intensity on T1-weighted images, whereas water increases signal intensity on T2-weighted images. Consequently, painful type 1 changes are associated with edema in the vertebral bodies. More recently, Jensen et al.[109] showed that these same Modic type 1 changes are strongly correlated with LBP. Albert et al.[110] proposed that anaerobic bacteria, predominantly *Propionibacterium acnes,* might be responsible for radiologic changes associated with LBP. The discs in their study infected with anerobic bacteria were significantly more likely to have Modic type 1 changes than discs infected with aerobic bacteria or discs with no detectable infection.[110] (Of note, whenever bacterial culture is taken from a surgical sample, there is a chance of environmental contamination, which could have affected these authors' results.) However, they found a difference between the aerobic and anaerobic bacteria groups. Simple environmental contamination would not likely favor anaerobic bacteria over the more commonly aerobic skin flora. Moreover, other studies have also found anaerobic bacteria in nucleus material, thus adding strength to the notion that the hypoxic conditions in the inner anulus and nucleus pulposus may be preferable for survival and colonization of anerobic microorganisms.[111,112]

The hypothesis that subclinical bacterial infection can cause LBP is further supported by the ability of antibiotic treatment to resolve back pain in patients with Modic type 1 changes. In a double-blind, randomized, controlled clinical trial, Albert et al.[113] found that a 100-day course of amoxicillin-clavulanate resulted in significant improvement in both disability and pain of patients with LBP. This is one of the most strikingly successful experimental treatments for LBP to date; however, along with the possible shortcomings of their earlier study, there are possible confounding effects of amoxicillin-clavulanate. There is some evidence suggesting that clavulanate has antiinflammatory action in the treatment of ulcerative colitis and analgesic properties during morphine withdrawal in mice, which could lead to a misinterpretation of this clinical study.[114,115] It is important to note that treating back pain with long-term antibiotics also raises global health concerns by exacerbating the

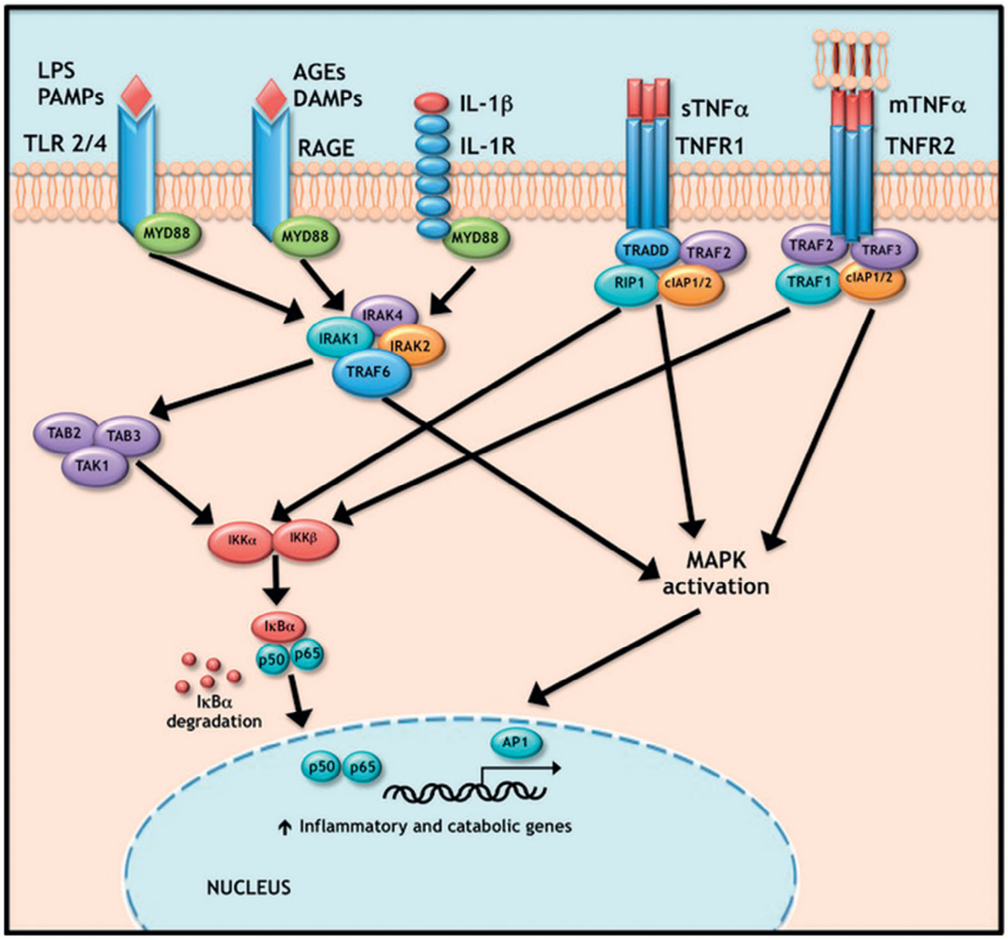

FIG. 46.3 Signaling pathways driving inflammation. Toll-like receptor (TLR), RAGE, and interleukin-1 (IL-1) all initiate their inflammatory effects through MYD88 with subsequent recruitment of members of the IRAK family that activate TRAF6. Tumor necrosis factor-α can stimulate the cell in both its soluble and membrane bound form by activating TNFR1 and TNFR2, respectively. TNFR1 ligand binding results in a conformational change leading to the recruitment of TRADD, RIP1, TRAF2, and CIAP1/2. Ligand binding to TNFR2 results in the recruitment of similar downstream factors. All five of these receptor pathways converge on nuclear factor (NF)-κB and mitogen-activated protein kinase (MAPK) activation to induce their inflammatory and catabolic effects. AGEs, advanced glycation end products; DAMP, damage-associated molecular patterns; LPS, lipopolysaccharide; PAMPs, pathogen-associated molecular patterns. (Reproduced from Gorth DJ, Shapiro IM, Risbud MV. Discovery of the drivers of inflammation induced chronic low back pain: from bacteria to diabetes. *Discov Med.* 2015;20[110]:177–184.)

already serious problem of antibiotic resistance.[116] Despite these limitations, subclinical bacterial infection is a promising new theory of LBP. Additional investigation into the underlying mechanisms linking anaerobic bacterial infection to LBP could enhance our collective understanding of the disease and identify treatments directly targeting the pathways contributing to painful disc degeneration.

Damage-Associated Molecular Patterns as Mediators of Chronic Inflammation and Low Back Pain

Some endogenous molecules and their atypical cleavage products have the ability to stimulate sterile inflammation in the disc. This family of molecules, collectively called damage-associated molecular patterns (DAMPs), include fragments of commonly found extracellular matrix molecules such as hyaluronan and fibronectin as well as high-mobility group protein B1 (HMGB1), a protein involved in chromatin assembly. Hyaluronic acid fragments (12–24 mer) have been reported to enhance the production of IL-1β, IL-6, IL-8, and matrix-degrading enzymes MMP-1, MMP-3, and MMP-13. These effects are mediated by Toll-like receptor-2 (TLR2) signaling and can be inhibited by high-molecular-weight hyaluronan.[117] The TLR2 signaling cascade activated by hyaluronan fragments is the same MyD88-IRAK-TRAF6–dependent cascade activated by both lipopolysaccharide and CpG sequence DNA binding (Fig. 46.3).[118,119] It is interesting to note that by binding TLR2 instead of TLR4 or CD44, hyaluronan fragments stimulate nucleus pulposus cells in a manner that more closely resembles macrophages than chondrocytes, hinting at the unique nature and origin of these cells.[120-122]

HMGB1 is another DAMP that is receiving attention in the disc field. HMGB1 is both passively released from necrotic cells and secreted by monocytes. It likely exerts its inflammatory function by binding to TLR2/4 as well as RAGE receptors.[122,123] It is important to note that redox status of HMGB1 is shown to play an important role in the biologic outcomes it produces.[124] Delivery of anti-HMGB1 reduces pain and TNF expression after application of nucleus pulposus tissue to DRG in a rat model.[125] Fibronectin fragments are another possible inflammatory mediator in disc degeneration. Fibronectin fragments are absent in infant human IVDs, but increase in concentration with aging and severity of disc degeneration.[126] Delivery of a 30-kDa fibronectin fragment into rabbit lumbar discs induced a degenerative phenotype similar to spontaneous human disc degeneration; it caused a significant decrease in disc height and proteoglycan synthesis.[127] These effects could be due to increased MMP-9 and MMP-13 expression along with decreased collagen II and aggrecan expression observed in vitro after stimulation with fibronectin fragments.[128] In contrast to signaling through RAGE and TLR, fibronectin fragments exert their catabolic effects through integrin $\alpha_5\beta_1$.[129] The enhanced catabolism caused by fibronectin fragments could also be a result of their outcompeting CCN2 for $\alpha_5\beta_1$ binding. CCN2 causes context-dependent effects in the disc, promoting aggrecan expression and inhibiting the catabolic effects of IL-1β when bound to $\alpha_5\beta_1$ and, conversely, possibly through HSPGs, promoting catabolism.[130] Thus, atypical matrix cleavage products may interfere with disc homeostasis, pushing the homeostatic balance more toward catabolism.

Systemic Metabolic Syndrome as a Driver of Disc Inflammation and Low Back Pain

Obesity and diabetes are two epidemics in Western society that increase the risk of developing painful disc degeneration. Beyond the greater forces that the spinal motion segments of an obese individual experiences, diabetes creates a niche environment that can drive disc degeneration. Advanced glycation end products (AGEs) associated with diabetes can accelerate disc degeneration by inducing catabolism and promoting inflammation. Although diabetes and obesity are linked, recent research has attempted to isolate the effects of diabetes from the effects of obesity by comparing both obese Sprague-Dawley rats to diabetic obese UCD-T2DM rats while using lean rats as an additional control. The diabetic rats alone had an elevated AGE concentration, which was associated with diminished glycosaminoglycan and water content of discs. These compositional changes resulted in an experimentally consistent decrease in disc mechanics.[131] In a separate study, Tsai and colleagues showed that treatment with AGEs of both rat and human nucleus pulposus cells caused a dose-dependent increase in MMP-2, extracellular signal-related kinases (ERK) activity, and elevation in advanced AGE-specific receptor (RAGE) mRNA expression, which all could lead to enhanced NF-κB activation.[132] Inhibiting RAGE-induced disc degeneration offers an accessible possible therapeutic target; both the antiinflammatory pentosan-polysulfate and AGE inhibitor pyridoxamine are already Food and Drug Administration (FDA) approved compounds and have been shown to reduce the deleterious effects of streptozotocin-induced diabetes on disc health in mice.[133] Independent of the aberrant reactions caused by hyperglycemia leading to AGE production, elevated blood glucose concentration, common to diabetes, has been shown to inhibit nucleus pulposus cell proliferation in vitro, which could further alter cellular homeostasis in the disc.[134] Although in vitro data and in vivo studies suggest that diabetes contributes to disc degeneration, the clinical data have not been as conclusive. A study evaluating nine pairs of monozygotic twins discordant for insulin-dependent diabetes mellitus did not show increased incidence of disc degeneration in the diabetic twin.[134a] A larger study focusing on non–insulin-dependent diabetes showed no increase in diabetic patient disc disease when controlled for body mass index.[134b] Taken together these studies suggest that diabetes, obesity, and their mechanical and metabolic consequences may compromise disc health and promote degeneration.

Effect of Disc Degeneration on Nerves

Degradative changes can occur within nerve roots exposed to nuclear material even in the absence of mechanical compression.[90,135-137] Weinstein et al.[138] investigated the reproduction of pain on discography and concluded that various neurochemical changes within the disc are expressed by sensitized anular nociceptors. These nociceptors are terminal nerve endings of sensory neurons that selectively respond to painful stimuli by the release of substance P.[139] These chemicals are leaked into the epidural space and transported into the axons of the exiting nerve roots. Within the nerve root they alter the excitability of type C nerve fibers and initiate the production of inflammatory agents such as prostaglandins as well as inflammatory cytokines, which results in radicular pain.[68,140,141]

In addition to material from the nucleus pulposus many other substances in the degenerated disc have been implicated in pain generation. The role of nitric acid and phospholipase A_2 in irritation of nerve roots has been well documented.[140,142-145] Phospholipase A_2 has been implicated in multiple aspects: direct activation of nociceptors, nerve injury from degradation of cell membrane phospholipids, and nerve injury from inflammatory mediators created from the arachidonic acid cascade (i.e., prostaglandins and leukotrienes).[146-148] Burke et al.[149] reported on the elevation of inflammatory mediators within the disc such as IL-6, IL-8, and prostaglandin E_2. Other studies have demonstrated the presence of inflammatory cytokines in the facet joints,[150] suggesting facet involvement of a pain generator via a biochemical mechanism as well. Ohtori et al.[151] reported on ingrowth of nerve tissue immunoreactive for TNF and neuronal marker protein gene peptide 9.5 (PGP) in 18 surgically harvested vertebral end plates of patients with Modic type 1 and 2 changes who had undergone surgery. Their findings suggest that axon ingrowth into the vertebral end plate in association with Modic changes was induced by TNF and may be related to pain generation.

Neurovascular proliferation within and around degenerated disc elements has been proposed as another mechanism of pain generation. Normal IVDs have sparse innervation and vascularity that is solely distributed within the outer lamellae (3 mm) of the anulus fibrosus,[152,153] whereas degenerated discs have significant neurovascular ingrowth within the inner anulus and nucleus pulposus,[154] although this observation remains a subject of controversy with the field. Immunoreactive staining and acetylcholinesterase studies have demonstrated penetration of nerve fibers within the inner third of the anulus in association with neovascularized granulation tissue.[142,152] Peng et al.[155] recently reported on a histological study of 19 IVDs harvested from surgery compared with normal control discs. The distinctive histologic characteristic of painful discs was a zone of richly innervated vascular granulation tissue extending from the outer anulus to the nucleus along the edges of fissures. Proliferation of vascular channels and sensory nerve endings rich in the calcitonin gene-related peptide (CGRP) has also been observed in the end plate region and vertebral body adjacent to the degenerated disc. These findings suggest a role for the vertebral end plate and body as additional pain generators in DDD.[156]

Other recent studies suggest that the sensory nerve supply within the IVD is similar to visceral innervation patterns,[157] with CGRP immunoreactive fibers that pass through the sympathetic trunks.[158] This visceral pattern of innervation is potentially susceptible to central sensitization, which may further complicate chronic LBP with psychosomatic overtones.[159] Psychosocial and chronic non–back pain syndromes have been implicated in recent publications as having a significant effect in patients with LBP.[160-164]

Clinical Picture

Internal Disc Disruption

The diagnosis of IDD is not readily apparent on a routine clinical workup. The patient is typically a younger individual between the ages of 20 and 50 years, with recurrent or persistent back pain. There may be a history of antecedent trauma or a forceful provocative event such as heavy lifting, or unexpected flexion or compression force on the lumbar spine, but more often the pain is gradual in onset with no associated event or date.

The pain is characterized as a deep, dull ache in the lower lumbar region, exacerbated by rotation, flexion, and side-bending movements and partially relieved by rest. Sitting intolerance may be a primary complaint, and pain is often relieved in a lateral recumbent position. Occasionally there is a complaint of pain in the buttock or posterior thigh, but there is a conspicuous lack of radiculopathic symptoms. In the rare instances of associated leg pain it is usually a late finding and pain does not follow any dermatomal pattern. O'Neill et al.,[165] in a study involving intradiscal electrothermal annuloplasty in 25 patients, showed that stimulation of the IVD may result in low back and referred leg pain in patients presenting with symptoms of IDD. The distal distribution of pain was found to depend on the intensity of stimulation, and occasional pain extending below the knee was produced.

Physical examination of these patients reveals decreased range of motion of the back and tenderness of the paraspinal musculature but is otherwise normal. The straight-leg raise test may reproduce back pain but not leg pain. LBP may also be reproduced at 20 to 30 degrees of flexion when rising from a flexed position. The sensorimotor examination is unremarkable, and deep tendon reflexes are normal and symmetric.

Degenerative Disc Disease

Patients with DDD typically present with a history of persistent LBP over the lumbosacral spine, sacroiliac joints, and radiating into the buttocks and posterior thighs. Symptoms are often exacerbated with sitting and prolonged walking; signs of neurologic claudication in the legs are not seen unless associated with concomitant lumbar stenosis. Radicular symptoms are rarely seen in the early stages of the disease. In end-stage DDD significant disc collapse may result in foraminal stenosis and late-onset radicular symptoms.

The physical examination is typically unremarkable except for point tenderness over the lumbar spine in the midline and over the sacroiliac joints. Range of motion of the lumbar spine may be reduced, most specifically in flexion. Extreme flexion usually causes significant discomfort as well as returning to upright from a flexed position. Extension is usually the least painful maneuver and may actually relieve pain. The straight-leg raise test may elicit some posterior thigh pain, which is often described as a stretching or pulling sensation, but there is no true radicular pain distal to the knee unless there is coexisting foraminal stenosis. The sensorimotor examination is usually unremarkable, and deep tendon reflexes are normal and symmetric.

Diagnostic Imaging

Plain Radiography

Plain radiographs are the recommended initial imaging modality for patients with LBP. Classic comparative and cost benefit studies have been performed to determine when and what radiographs to obtain.[166,167] In 1982 Liang et al.[168] published a comparison study between performing radiographs on all patients versus only on patients whose pain did not improve within 8 weeks of presentation. They found that risks and costs did not justify obtaining radiographs on initial presentation. Scavone et al.[169] reviewed the radiographs of 782 patients and found that spot lateral and oblique films only added diagnostic information in 2% of patients. They recommended that a spine series in patients with LBP should consist only of anteroposterior and lateral films. In general, flexion-extension and oblique views are only necessary in patients suspected of having instability or a pars fracture. The presence of so-called red flags increase the chances of diagnostic radiographic findings and may prompt the physician to obtain early radiographic studies (Box 46.1).[170]

Typical radiographic findings for patients with DDD include narrowing of the disc space (loss of height), end plate

BOX 46.1 Practice Guidelines: Red Flags

Anteroposterior and lateral radiographs are generally not useful in the acute setting but may be warranted with the following:

- Unrelenting night pain or pain at rest (increased incidence of clinically significant pathology)
- History or suspicion of cancer (rule out metastatic disease)
- Fever above 38°C (100.4°F) for >48 hours
- Osteoporosis
- Other systemic diseases
- Neuromotor or sensory deficit
- Chronic oral steroid administration
- Immunosuppression
- Serious accident or injury (fall from heights, blunt trauma, motor vehicle accident), not including twisting or lifting injury unless other risk factors are present (e.g., history of osteoporosis)
- Clinical suspicion of ankylosing spondylitis

Other conditions that may warrant anteroposterior or lateral radiographs:

- Age >50 years (increased risk of malignancy, compression fracture)
- Failure to respond to 4 to 6 weeks of conservative therapy
- Drug or alcohol abuse (increased incidence of osteomyelitis, trauma, fracture)

Oblique radiographs are not recommended; they add only minimal information in a small percentage of cases and more than double the exposure to radiation.

From Institute for Clinical Systems Improvement (ICSI). *Adult low back pain.* Bloomington, MN: ICSI; 2008.

sclerosis, and the presence of osteophytes. These are, however, late changes occurring when the degeneration is advanced. Degenerative spondylolisthesis and scoliosis may occur as degeneration progresses. Advanced stages of disc degeneration may demonstrate vacuum phenomenon within the discs, a finding that represents nitrogen collection within voids in the disc.

The radiographs in patients with IDD typically show well-preserved height in the IVD and are normal in appearance other than occasional benign spinal alignment changes. Nonstructural scoliosis and loss of lumbar lordosis may safely be observed in patients with sciatic list and paraspinal spasm.

Computed Tomography

Computed tomography (CT) is an excellent method to assess osseous pathology, but it is generally not the imaging modality of choice for either IDD or DDD because they are primarily soft tissue disorders. Injecting contrast material into the vertebral canal (CT myelography) significantly improves the accuracy of the CT scan for demonstrating pathology within the canal such as masses or stenosis; although they are not a primary feature of DDD, they frequently occur secondarily. CT myelography is the diagnostic imaging study of choice in patients with significant scoliosis or those who are unable to undergo MRI examination due to implanted metal, aneurysm clips, pacemaker, obesity, or claustrophobia.

CT scanning has largely been replaced by MRI scanning because the disc is not adequately imaged, and CT exposes the patient to radiation. As is discussed later, CT is also used in combination with discography.

Magnetic Resonance Imaging

T1- and T2-weighted MRI is used in standard protocols. A semiquantitative method is often used to classify degenerative grades. The 5-scale grading system by Pfirrman combines structural changes with distinction of nucleus and anulus, signal intensity, and height of the disc. Quantitative MRI techniques such as relaxation times and T1ρ have the potential to biochemically assess disc tissue but are currently rarely used in clinical practice. Diffusion of water protons and contrast agents are other quantitative methods primarily used in research. MRI spectroscopy is challenging in vivo but may in the future provide important information about the source of pain.

MRI is the best imaging modality for visualization and evaluation of the neuronal and discal elements and is the most valuable adjunctive diagnostic tool in assessing disc pathology. The IVDs are a highly unlikely cause of pain if the patient's MRI is completely normal and all discs are well hydrated. General MRI findings indicative of DDD include loss of water, loss of disc height, disc bulges, and signal or morphologic irregularity within the nucleus pulposus and end plates. In addition, MRIs are often examined for three specific types of findings: a high-intensity zone (HIZ) in the posterior anulus, dark disc with or without loss of height, and end plate signal changes.

The MRI finding of an HIZ was originally described by Aprill and Bogduk[171] in 1992 and is believed to represent an anular tear (see Fig. 46.1). Postmortem studies have demonstrated three types of tears that can occur in the anulus: concentric, transverse, and radial.[172,173] A concentric tear is a crescentic or oval cavity created by a disruption in the short transverse fibers interconnecting the anular lamellae and is usually not visible on MRI. These concentric tears are occasionally referred to as *delamination*. A transverse tear represents a rupture of the Sharpey's fibers near their attachments to the ring apophysis at the disc periphery; these tears are typically thought to be clinically insignificant. A radial tear extending from the nucleus pulposus to the outermost surface of the posterior anulus is manifest on MRI as an HIZ.[174] The HIZ is visualized on T2-weighted spin-echo images as high-intensity signal located within the anulus fibrosus that is clearly distinguishable from the nucleus pulposus.

Decreased signal within the IVD on T2-weighted images with relative preservation of disc height is a relatively common finding in asymptomatic individuals. Such a disc appearance is frequently referred to as *dark disc disease*; however, whether these discs constitute a potential pain generator is unclear. In the absence of any psychometric abnormalities, an isolated dark disc in a patient with no other identifiable causes of back pain is considered by many clinicians to be a source of back pain even though the evidence is weak to absent.

End plate changes (see Fig. 46.2) that occur with disc degeneration have been well described by Modic.[175] Stage 1 change represents edema and is characterized by decreased signal on T1-weighted images and bright signal on T2-weighted images within the end plate. In stage 2, fatty degeneration in the bone adjacent to the end plates is represented by bright signal on T1-weighted images and intermediate signal on T2-weighted images. Lastly, stage 3 changes correspond with advanced degenerative changes and end plate sclerosis and are characterized on MRI by decreased signal intensity on both T1- and T2-weighted images.

When interpreting MRI findings, the clinician must be careful to consider the high prevalence of clinically false-positive findings. Abnormal disc findings on MRI are often found in clinically asymptomatic individuals. Boden et al.[176] showed that approximately 30% of asymptomatic individuals have a major finding on lumbar MRIs. In patients older than 60 years those abnormal findings are almost universally present irrespective of symptoms. Jensen et al.[177] reported on 98 asymptomatic patients aged 20 to 80 years and found that 52% overall had disc bulge in at least one level on MRI. Stadnik et al.[178] showed an unusually high rate of disc bulge (81%) and anular tears (56%) on MRI in their 30 asymptomatic volunteers.

Abnormal MRI findings in asymptomatic patients are not indicators of future problems. Borenstein et al.[179] reported on 50 of the 67 patients from the Boden study at a 7-year follow-up interval and found that incidental MRI findings were not predictive of the development or duration of LBP. Jarvik et al.[180] studied 148 veterans who had been asymptomatic of LBP for at least 4 months. They found an incidence of moderate to severe desiccation in at least one disc in 83%, disc bulge in 64%, and loss of disc height in 58%. In a 3-year follow-up of the same cohort, the investigators found no association between the development of new back pain and incidental MRI findings such as Modic changes, disc degeneration, anular tears, or facet degeneration. The single greatest risk factor for developing LBP in the 3-year interval was depression.[160]

Jarvik et al.[181] also published a report on the use of early MRI in the primary care setting. A total of 380 patients with LBP were randomized to receive either initial spine imaging via rapid MRI or plain radiography. These authors reported that substituting rapid MRI for radiographic studies in the primary care setting offered little additional benefit to patients in terms of secondary outcomes measures at 1 year and had the potential to increase the cost of care by $320 per patient (in 2002 dollars). Carragee et al.[182] performed a prospective study of 200 asymptomatic patients to determine the rate at which new episodes of LBP are associated with changes on MRI. On follow-up MRI in 51 patients who had an episode of LBP, 84% had no new finding. The most common new findings were disc signal loss (dark disc), progressive facet arthrosis, and increased end plate changes. New findings were not more common in patients developing back pain after minor trauma. The conclusion was that new findings on MRI within 12 weeks of onset of a serious episode of LBP were unlikely to represent any significant structural change and preexisted the onset.

In consideration of the high prevalence of false-positive MRI findings, the clinician does well to remember that the MRI does not stand alone in the evaluation of spinal pathology. Together with the patient's history, physical findings, and plain radiographs, selective use of MRI can provide valuable information on the source of a patient's lumbar complaints.

Contrast-Enhanced Magnetic Resonance Imaging

The use of intravenous gadolinium diethylenetriaminepenta-acetic acid (DTPA) contrast with MRI in the setting of discogenic pain has also been explored. The addition of gadolinium to a lumbar MRI is known to be useful for differentiating scar tissue from recurrent disc herniation, as the latter fails to enhance while the vascular scar tissue will take up the contrast. It appears unlikely that gadolinium-enhanced MRI helps determine if a degenerative disc is painful. Lappalainen et al.,[183] in an animal study of surgically created anular tears, showed that gadolinium-enhanced MRI did not detect all tears; specifically the peripheral, small tears were not visualized. Yoshida et al.[184] investigated the relationship between T2-weighted gadolinium-DTPA–enhanced MRI and a positive pain response with discography of 56 lumbar discs in 23 patients with chronic LBP. The sensitivity, specificity, positive predictive value, and negative predictive value of the unenhanced T2-weighted images in detecting the symptomatic disc were 94%, 71%, 59%, and 97%, respectively. With enhancement the corresponding values were 71%, 75%, 56%, and 86%, respectively.

High-Intensity Zone

The presence of an HIZ indicates the presence of an anular fissure. Aprill and Bogduk[185] correlated the finding of an HIZ with CT discography and found an 86% positive predictive value for a positive discogram; however, the predictive value and clinical significance of the HIZ on MRI has more recently been brought into question. The results vary among publications. Multiple authors have found a positive correlation between the finding of an HIZ and concordant pain on discography similar to the findings of Aprill and Bogduk,[186-189] whereas others have documented the correlation but found unacceptably low sensitivity.[190,191]

Kang et al.,[192] in a study of 62 patients aged 17 to 68 years, found that only an HIZ in association with disc protrusion correlated with concordant pain on discography. The specificity was 98% and positive predictive value was 87%; however, the sensitivity was low at 46%. An HIZ in association with either a normal or bulging disc on MRI was not found to be associated with a positive discogram. Ricketson et al.,[193] in a 30-patient study, found no correlation between the presence of an HIZ on MRI and a concordant pain response on discography; however, they did note that an HIZ was never visualized in a disc found to be morphologically normal on discography. Further studies[144,187,194-196] attempting to correlate positive HIZ findings on MRI and painful discography suggest that, although lumbar IVDs with posterior combined anular tears are likely to produce pain, the validity of these signs for predicting discogenic lumbar pain is limited.

Although the prevalence is unknown, an HIZ is frequently seen in asymptomatic individuals.[139] Carragee et al.[162] reported the presence of an HIZ in 59% of their symptomatic patient population and in 24% of asymptomatic individuals. In the asymptomatic group, 69% of the discs with an HIZ were positive on discography, whereas 10% of the discs without an HIZ were positive. The authors also reported that 50% of the discs with an HIZ were positive on discography in patients with normal psychometric testing compared with 100% positive discography results in patients with abnormal psychometric testing and chronic pain. They concluded that the presence of an HIZ does not reliably indicate the presence of symptomatic IDD due to the high prevalence of the HIZ in asymptomatic people.

In 2004 Mitra et al.[197] followed 56 LBP patients with the finding of an HIZ longitudinally for 6 to 72 months with MRI. Changes in the HIZ on follow-up MRI—either an increase in intensity or spontaneous resolution—were not correlated to changes in Visual Analog Scale (VAS) score, Oswestry Disability Index (ODI), or symptoms, which again calls into question the clinical significance of the HIZ. Although the finding of an HIZ on MRI has been found in some studies to have good specificity and positive predictive value for concordant pain generation on discography, it has low sensitivity, high false-positive rates, and therefore questionable clinical significance.

Dark Disc

It is quite clear that the majority of patients with a dark disc are asymptomatic. Dark disc is, however, clearly the sign of a degenerative process. Milette et al.[198] found that loss of disc height or abnormal signal intensity were highly predictive of symptomatic tears extending beyond the anulus. Horton and Daftari[199] reported a positive discogram in 50% of patients with dark discs without evidence of an anular tear. An isolated dark disc with concordant pain on provocative discography is often considered to be pathologic in the absence of other potential sources of pain and in the absence of confounding psychosocial issues; however, as discussed previously, this evidence is weak.

Modic End Plate Changes

The various stages of Modic change are thought to be specifically linked with phases of the degenerative disc process. Toyone et al.[200] evaluated the MRIs of 74 patients with Modic changes and found that type 1 changes tended to be associated with complaint of LBP and correlated to segmental hypermobility. Other investigators have also described Modic 1 changes as specifically associated with LBP.[201,202] In a recent large retrospective review by Thompson et al.,[203] Modic changes in 736 patients were correlated to provocative discogram. The authors found that Modic type 1 changes had a high positive predictive value (0.81) for a positive discogram. Modic type 2 had a lower positive predictive value (0.64), and the predictive value of Modic type 3 was not statistically significant.

In the original Modic et al.[175] description of vertebral body marrow changes, conversion between signal characteristics from type 1 to type 2 was described in five of six patients over the course of 14 months to 3 years. Mitra et al.[204] performed a prospective evaluation of 48 patients with Modic type 1 changes. At 12-month to 3-year follow-up, 37% were found to have progressed to Modic type 2, 15% partially progressed, and 40% had no change in type but more extensive Modic 1 changes. Type 1 changes are believed by many to represent an unstable, dynamic phase of the degenerative process and tend to either convert to a type 2 pattern or become more pervasive. Modic type 2 changes are thought to be stable and less associated with painful episodes, but there have been reports of type 2 converting back to type 1.[205] Kuisma et al.[206] reported the prevelance of Modic changes in 60 patients treated nonoperatively for sciatica to be 23%. In a longitudinal follow-up of the same patients at 3 years, 14% were noted to have changed type. Those levels that did not convert to type 2 were found to have more extensive type 1 changes. Development of a Modic change at previously unaffected levels was found in 6%.

Many authors have explored the correlation between Modic changes on MRI with positive concordant pain on discography. Sandhu et al.[207] found that while both were relatively specific for discogenic pain, there was no significant correlation between them. Braithwaite et al.[208] found that Modic changes did not predict a positive response on discography; they concluded that Modic changes may represent a specific but relatively insensitive sign of discogenic LBP. Kokkonen et al.[209] observed that contrast injection during discography reflected pain of discogenic origin well, whereas the pain associated with end plate damage was usually not demonstrated by CT discography. The authors found a stronger association between end plate degeneration and disc degeneration than between end plate degeneration and anular tears, which may explain why Modic changes have been found to be less sensitive for discogenic pain than discography.

Conversely, other studies have found better correlation between back pain and Modic changes than the correlation between back pain and discography. Carragee et al.[161] reported on 100 prospectively followed patients recruited from a study population of asymptomatic persons at high risk for developing disabling back pain. Of all the incidental diagnostic findings, only moderate or severe Modic changes of the vertebral end plates were found to be weakly associated with subsequent development of a disabling episode of back pain. Other structural MRI findings as well as concordant pain with discography correlated only weakly with previous back pain episodes and had no association with future disability or medical consultations for back pain. Interestingly, psychosocial, neurophysiologic (chronic nonlumbar pain), and occupational factors strongly predicted future disabling episodes and consultations for back pain.

Schenk et al.,[210] in a cross-sectional study of 109 women from two groups (nursing or administrative professions), found that Modic changes and nerve root compromise were the only MRI findings that were statistically significant predictors of LBP. Disc degeneration, disc herniation, HIZs, and facet arthritis were found in both groups but were not significant risk factors for LBP.

Similar findings were reported in a study by Kjaer et al.,[211] in which LBP was correlated to MRI findings in a random selection of 412 Danish subjects. Although Modic changes occurred in less than 25% of subjects (16% Modic type 1 and 7% Modic type 2), this finding had the strongest correlation with complaints of back pain. When the subjects were evaluated clinically, the authors found that patients with both radiographic evidence of DDD and Modic changes had the best clinical evidence of disc disease. Clinical findings in patients with radiographic evidence of disc degeneration without Modic changes were not significantly different from the baseline population. The authors concluded that a Modic change was a critical finding in relation to history of LBP and clinical findings.[212] In a follow-up study of the same Danish population, Modic changes correlated with type of occupation, history of smoking, and overweight body habitus. The odds ratio for heavy labor combined with smoking was 4.9 for the presence of Modic changes on MRI.[213]

A recent meta-analysis review of Modic changes by Jensen et al.[214] found that the median prevalence of Modic changes from all studies was 43% in patients with nonspecific LBP. Seven of 10 studies reported a positive association between LBP and Modic changes with odds ratios between 2.0 and 19.9.

Axially and Vertically Loaded Magnetic Resonance Imaging

Axially loaded MRIs have been used to evaluate patients afflicted with lumbar spinal diseases. The idea is to better reproduce the anatomy of the disc under physiologic load. The utility of axially loaded MRI has been primarily studied in the spinal stenosis and spondylolisthesis patient populations.[215-218] Danielson and Willén[219] observed a significant decrease in dural cross-sectional area between a psoas-relaxed position and axial compression in extension in 56% of asymptomatic individuals. The decrease was most pronounced at L4–L5 and greater in older individuals. Saifuddin et al.[220] postulated that lumbar spine MRI with axial loading may increase the sensitivity for the detection of HIZs; however, this hypothesis remains untested.Vertical MRI has also been introduced to represent a loaded spine situation, but no evidence suggests a role in pure DDD.

Discography

Discography is another diagnostic modality for the evaluation of the integrity of the lumbar disc. There is significant controversy surrounding its usefulness. Some investigators consider discography to be the single most important tool in the diagnosis of IDD,[138,221] but recent outcome studies[222] and a practice guideline by the American Pain Society[223] have recommended against the use of provocative discography in the diagnosis of discogenic back pain.

Discography is the only widely available physiologic modality used to determine if a specific disc is a pain generator. Although several attempts have been made to explain the pathogenesis of pain provocation during discography, the precise pathomechanism is not well understood. There are four components to the evaluation of a discogram: (1) the pressure and volume of fluid injected into the disc, (2) the morphology of the disc being injected, (3) the subjective pain response at the level of interest, and (4) the pain response when adjacent control levels are injected.[224,225] The subjective pain response to low-pressure provocation is believed to be the most important determinant of disc derangement; reproduction of the patient's symptoms upon injection of the diseased level is essential to a positive test. A normal disc nucleus can accept 1.0 to 1.5 mL of contrast media. If 2.0 mL or more of contrast is easily introduced, some degree of disc degeneration is assumed with contrast leakage from the nucleus. The use of postdiscography CT increases the sensitivity for the diagnosis of radial tears of the anulus[226]; however, because of low specificity and sensitivity, postdiscography CT is not as helpful in the diagnosis of IDD. Most authors feel that to be diagnostic the pain should be concordant on low-pressure injection and a normal control disc should be pain free.

Despite being used since 1948, discography remains controversial. In the 1960s, Holt et al.[221] and Massie et al.[227] reported on the high false-positive rate of lumbar discography, which was found to be as high as 26% by Holt. Walsh et al.[228] later published a rigorous study on the reliability of lumbar discography. Ten normal volunteers were compared with seven symptomatic patients. Although 17% of the normal discs were found to be morphologically abnormal, there were no positive pain responses. They concluded that with modern techniques the false-positive rate of lumbar discography is not as high as reported by Holt. Derby et al.[229] found similar results in a more recent study of 90 LBP patients and 16 controls. Morphologically, the prevalence of grade III anular tears was 58% among the asymptomatic control population. Presumably asymptomatic discs in symptomatic individuals on pressure-controlled discography demonstrated pain levels and responses similar to the control group, whereas patients with true-positive discography demonstrated pain characteristics concordant with their usual symptoms. The authors concluded that pressure-controlled discography can differentiate between asymptomatic discs and morphologically abnormal discs.

Carragee et al.[163] studied the false-positive rate of low-pressure discography in a comparison of 69 volunteers with no significant LBP and 52 patients undergoing discography in consideration for treatment of discogenic pain. Low-pressure discography was positive in at least one level in 27% of the LBP patients and in 25% of the controls. The false-positive rate of discography was 25% and correlated with psychosocial factors and history of chronic pain of a non–lumbar spine origin. In another publication from Carragee's group,[164] psychosocial factors and chronic nonlumbar pain, such as cervical pain and somatization disorder, also correlated with positive discography in patients asymptomatic for LBP. The authors concluded that false-positive rates can be low with strict application of the Walsh protocol[228] in patients who do not have positive psychometric issues or other chronic pain syndromes.

In contrast to reports of high false-positive rates, two recent meta-analyses of low-pressure discography reported strong evidence to support the role of discography in identifying patients with discogenic pain.[230,231] Combined data from all studies demonstrated an overall false-positive rate of 9.3% per patient and 6.0% per disc. False-positive rates among asymptomatic patients was 3.0% per patient and 2.1% per disc. Chronic pain was not found to be a confounder, and strength of evidence was reported as level 2 in support of the diagnostic accuracy of discography.

Finding a gold standard to which discography results can be compared remains a problem. Few studies have compared the use of discography and outcomes after surgical fusion, which is perhaps the best measure for the validity of discography. Colhoun et al.,[232] in a study of 137 patients, reported 89% favorable outcomes in patients with positive concordant pain on discography versus 52% favorable outcomes among those who had no painful response. Madan et al.[233] had different findings; 81% of 41 patients who underwent fusion based on MRI findings had satisfactory outcomes versus 76% of 32 patients who had surgery based on discography. Perhaps the most rigorous study to date was published by Carragee et al.[234] Success of surgical fusion was compared in 32 patients with single-level positive discogram and a matched cohort of 34 patients with single-level spondylolisthesis. Seventy-two percent of the spondylolisthesis patients met the highly effective success criteria for surgery versus only 27% of patients with discogenic pain. Minimal acceptable success criteria were 91% and 43%, respectively. The authors calculated a best-case positive predictive value for discography of 50% to 60% and concluded that provocative discography was not highly predictive of single-level discogenic back pain.

In an attempt to improve on the poor reliability of discography, interest has turned to functional anesthetic discograms, also called discoblocks. This is a modification of discography in which a local anesthetic, usually bupivacaine, is infused with the contrast agent into the disc to enhance the diagnostic capability of the procedure. Relief of pain after discoblock is considered diagnostic for discogenic pain. A recent randomized controlled study comparing standard provocative discogram to discoblock in diagnosing discogenic LBP was published by Ohtori et al.[235] Anterior lumbar interbody fusion procedures were performed in 15 patients who were diagnosed with discogenic pain by discography and 15 patients diagnosed by discoblock. Outcome measures (ODI, VAS, and Japanese Orthopedic Association score) at 3 years' follow-up showed statistically significantly better results in the group diagnosed by discoblock.

Regardless of the details of how discography is performed, some authors have posed the question of potential ill effects resulting from perforating the lumbar disc. Carragee et al.[222] recently published a report on the effect of lumbar discography in precipitating accelerated degeneration in a matched cohort study. Ten-year follow-up demonstrated that discs that had been punctured had a greater progression of disc degeneration (35% vs. 14% in the control group). There were 55 new disc herniations in the discography group versus 22 in the control group. The authors concluded that despite utilizing modern discography techniques with small-gauge needles, there is still an increased risk of disc degeneration, disc herniation, changes in disc and end plate signal, and loss of disc height when discography is performed.

Although discography has the potential for diagnosing disc derangement, its reliance on the patient's subjective pain response can also be problematic when secondary gain may be an issue. Psychosocial factors and chronic nonlumbar pain have also been shown to alter the diagnostic capabilities of the procedure. Finally, consideration of the consistent reports of the high false-positive rates, as well as new findings of accelerated degeneration in discs that undergo discography, make it difficult to recommend the procedure for the diagnosis of discogenic back pain. Indeed, the validity of lumbar discography is very much in doubt, which is underscored by a recent practice recommendation published by the American Pain Society. The society's current recommendation is that provocative lumbar discography should not be used for making the diagnosis of a discogenic source of pain in the setting of nonradicular LBP.[223] The value of using discography to assess the levels to be operated on in patients with multilevel disc degeneration has not been adequately established scientifically.

Treatment

Once a clinician has gathered all the data from the history and physical examination along with appropriate diagnostic studies, decisions must be made regarding treatment. All available information should be used in formulating a treatment plan to ensure a successful outcome. Sole reliance on individual clinical findings or imaging studies drastically lowers the success rate because the incidence of disc abnormality in asymptomatic patients approaches 30% to 40% and increases with advancing age.

In 2009, the American Pain Society published five practice guidelines regarding the management of chronic nonradicular back pain based on the best available evidence for the various diagnostic and treatment modalities available. Those recommendations are summarized in Box 46.2.[223] These treatment modalities, along with others not mentioned in the treatment recommendations, are discussed in detail along with brief summaries of the current supporting and opposing literature.

Nonoperative Treatment

Nonoperative treatment of lumbar disc disorders has been extensively discussed in the literature.[236,237] Physical therapy, pharmcology, and spinal manipulations have all been supported by mutliple studies of reasonable quality, but it is difficult to fully evaluate most of these studies because of a generalized lack of randomized control design, blind observers, compliance measures, and cointerventions. Additionally, very little of the literature on these nonoperative treatments is specific for the diagnosis of IDD or DDD, but rather is generalized to chronic and acute LBP, which may have multiple etiologies.

BOX 46.2 American Pain Society Recommendations Regarding Management of Chronic Nonradicular Low Back Pain

Recommendation 1

- Strong recommendation against use of provocative discography as a procedure for diagnosing discogenic low back pain (moderate-quality evidence).
- Insufficient evidence to evaluate validity of diagnostic selective nerve root block, facet joint block, medial branch block, or sacroiliac joint block as diagnostic procedures.

Recommendation 2

- In patients who do not respond to usual, noninterdisciplinary interventions, clinicians should consider intensive interdisciplinary rehabilitation with a cognitive/behavioral emphasis (high-quality evidence).
- Clinicians should counsel patients about interdisciplinary rehabilitation as an initial treatment option.

Recommendation 3

- Facet joint corticosteroid injection, prolotherapy, and intradiscal corticosteroid injection are not recommended (moderate-quality evidence).
- Insufficient evidence to adequately evaluate benefits of local injections, botulinum toxin injection, epidural steroid injection, intradiscal electrothermal therapy, therapeutic medial branch block, radiofrequency denervation, sacroiliac joint steroid injection, or intrathecal therapy with opioids or other medications.

Recommendation 4

- Clinicians should discuss risks and benefits of surgery including a specific discussion about intensive interdisciplinary rehabilitation as a similarly effective option, the small to moderate average benefit of surgery over noninterdisciplinary nonsurgical therapy, and the fact that the majority of patients who undergo surgery do not experience an optimal outcome (moderate-quality evidence).

Recommendation 5

- Insufficient evidence to adequately evaluate long-term benefits and harms of vertebral disc replacement.

From Chou R, et al. Interventional therapies, surgery, and interdisciplinary rehabilitation for low back pain: an evidence-based clinical practice guideline from the American Pain Society. *Spine.* 2009;34:1066–1077.

Bed Rest and Advice to Stay Active

The use of bed rest and its duration has long been debated in the literature. Treatment schedules ranging from 2 days to 6 weeks have been described.[238-240] The currently accepted recommendation[170] is limited bed rest for a maximum of 2 days in patients with acute severe pain because longer durations of bed rest may be detrimental to the patient's general health while offering no benefit to the back pain. Even a short period of bed rest may not be beneficial. Allen et al.[241] published a review of studies of bed rest as treatment for 15 different conditions and found that for patients with acute LBP there was significant worsening of outcome measures. A Cochrane Review of bed rest for treatment of acute LBP reported high-quality evidence that advice to rest in bed is less effective than advice to stay active.[242] Progressive return to activity and the initiation of a formal physical therapy or home exercise program are recommended after any initial short period of rest.

Verbunt et al.[243] explored reasons why patients sometimes use prolonged bed rest in the setting of acute episodes of LBP. Among the study population of 282 patients, 33% reported using bed rest, and 8% remained in bed for longer than 4 days. Behavioral factors, catastrophizing, and fear of injury were associated with use of prolonged bed rest. History of back pain and pain intensity was not associated with patient use of prolonged bed rest. Additionally, patients who used prolonged bed rest in the early phase of acute LBP were more disabled after 1 year.

Patient education and advice to stay active is now the favored recommendation. A Cochrane Review[244] of patient education and advice to stay active demonstrated strong evidence that individual instructional sessions of 2.5 hours are more effective in returning patients to work than no intervention; however, in the setting of chronic back pain, patient education was less effective than more intensive interventions. Education sessions of shorter duration or written information was no more effective than no intervention. Another meta-analysis[245] of 39 randomized controlled studies evaluated whether advice to stay active alone was as effective as advice in combination with other interventions such as "back school" or specific exercise routines. Advice as an adjunct to a specific exercise program was the most common form of treatment in the reviewed studies and was best supported for chronic LBP. Outcomes among acute LBP patients were generally poor, but advice to stay active alone was found to be the best recommendation.

Brox et al.[246] published a systematic review of brief education in the clinical setting involving examination, information, reassurance, and advice to stay active. The authors found strong evidence that brief education was more effective for return to work but was no more effective than usual care for reduction of pain. There was limited evidence that exercise was more effective than dissemination of a back book or an educational internet session.

Orthotics

Another common conservative management technique involves the use of lumbar supports. Calmels et al.,[247] based on the results of a randomized clinical trial, recommended the limited use of a lumbar belt to improve functional status and pain and reduce medication use. Oleske et al.[248] performed a randomized clinical trial of back supports and patient education in work-related back pain. The authors found no effect on patient self-report of recovery or lost work time between brace use and controls, but back supports were found to have value in preventing recurrence of work-related back pain. A Cochrane systematic review[249] of brace treatment for LBP failed to find sufficient evidence to support the use of lumbar supports to treat LBP. Moderate evidence was found that braces are no more effective than no treatment or physical training in preventing episodes of back pain.

Shoe insoles have been recommended in the past for both treatment and prevention of nonspecific LBP. The most recent Cochrane systematic review[250] of six randomized control trials reported strong evidence that use of insoles does not prevent episodes of LBP. Limited evidence was found that insoles alleviated LBP, but no conclusions or recommendations were made for use in the treatment of patients with LBP.

Physical Therapy

A large number of physical therapy modalities and routines have been described in the literature. They include land-based and aquatic programs, specific protocols and exercise routines, and group treatment programs—so-called *back schools*. Adjunctive modalities include ultrasound, iontophoresis, transcutaneous electrical nerve stimulation, and heat and cold therapy. Exercise programs commonly use aerobic exercise, stretching, flexion and extension routines, core conditioning, and back stabilization protocols. The goal is to improve core strength, flexibility of the trunk and hip muscles, and conditioning. Patients often respond differently to physical therapy so treatment programs must be tailored to the individual. Periods of activity modification may be necessary. Patients should also be educated on proper body biomechanics, lifestyle, and healthy living habits such as weight control, proper nutrition, stress relaxation, and cessation of smoking.

Several published randomized controlled trials have supported different therapy routines or programs. Although a comprehensive review of all the various programs is not in the scope of this chapter, some critical updates in recent years are worthy of discussion. Recent prospective randomized trials comparing physical therapy to fusion have emphasized the importance of a multidisciplinary approach with cognitive therapy, fear-avoidance counseling, and intensive exercise programs.[251-253] A recent systematic review[246] found moderate evidence that fear-avoidance training emphasizing exposure is more effective than graded activity increase for improvement of pain, disability, and fear avoidance.

Intensive interdisciplinary rehabilitation with emphasis on cognitive and behavioral intervention is one of the treatment recommendations by the American Pain Society.[223] Interdisciplinary rehabilitation was defined as an integrated intervention with rehabilitation and a psychological and/or social or occupational component. Noninterdisciplinary or "traditional" physical therapy is also efficacious in this patient population, but no one specific program, method, or technique is significantly better than another. A recent study of 3094 patients treated with physical therapy for back pain concluded that the treatment is safe but did not reach minimally effective clinical level.[254]

Back schools are another commonly discussed therapy modality, and there is some indication that low-intensity back schools may have some efficacy. Heymans et al.,[255] in a randomized controlled trial, found that low-intensity back school patients experienced fewer sick leave days (68 days) than usual care patients (75 days) and those who attended high-intensity back school (85 days). Functional status and kinesiophobia were improved at 3 months, but there was no difference in pain intensity and perception of recovery among the groups. In another randomized controlled trial, however, Kaapa et al.[256] found no significant benefit of back school compared with physical therapy combined with cognitive therapy at 6-month, 1-year, and 2-year follow-up.

A 2004 systematic Cochrane Review[257] concluded there was moderate evidence suggesting that back schools in an occupational setting reduce pain and improve function and return-to-work status compared with other forms of therapy such as exercises, manipulation, myofascial therapy, advice, placebo, and waiting list controls. Brox et al.[246] published a separate systematic review of back schools and found moderate evidence that back schools were no better than waiting lists, no intervention, placebo, or general exercises for reduction of pain.

A European economic evaluation of a randomized controlled study[258] of intensive group therapy found no significant cost difference between intensive group therapy and standard physiotherapy. There was also no difference in clinical effect between the groups at 1 year follow-up.[259] To date there are no economic studies of group therapy back schools in the United States. Although low-intensity back school and programs in a work setting may have a benefit versus other forms of nonoperative treatment, most of the current literature demonstrates that back schools offer little benefit over standard physiotherapy and cognitive therapy.

Adjunctive Modalities

Another treatment option for LBP includes adjunctive physical therapy modalities such as transcutaneous electrical nerve stimulation (TENS), electrical muscle stimulation, ultrasound, and iontophoresis. Poitras and Brosseau[260] published a review of randomized controlled data on the use of TENS and found that it may be useful for immediate short-term pain reduction but has little impact on patient perception of disability or long-term pain control. A 2008 Cochrane Review of TENS versus placebo[261] concluded there is currently not enough evidence to support the routine use of TENS for the management of chronic LBP.[262] There is even less literature available on the use of iontophoresis and ultrasound in the setting of discogenic back pain. The few randomized controlled trials that exist focus on ultrasound in conjunction with other physical therapy regimens. The efficacy of these modalities as well as heat and cold treatments in isolation has not been determined.

Manual Therapy and Complementary/Alternative Medicine Therapies

Several studies have reported beneficial effects of chiropractic treatment for acute nonspecific LBP.[263-265] The role of chiropractic manipulations for the treatment of IDD or DDD of the lumbar spine has not been studied. Chiropractic manipulation is generally not considered effective in the treatment of chronic back pain resulting from disorders of the IVDs.[266] A Cochrane Review[267] reported there was no evidence that spinal manipulative therapy was superior to general practioner care, analgesics, physical therapy, exercises, or back school in the treatment of both acute and chronic LBP.

Eisenberg et al.[268] published a randomized trial of usual care therapy (nonsteroidal antiinflammatory drugs [NSAIDs], educational and activity modification) versus the addition of the patient's choice of alternative therapy—chiropracty, acupuncture, or therapeutic massage—in the treatment of acute LBP. Outcomes using the Roland-Morris scale and subjective assessment of symptoms demonstrated no statistically significant improvement in patients who underwent alternative therapies. The study did, however, demonstrate an increase in patient satisfaction with alternative therapy.

Hurwitz et al.[269] had similar findings in a randomized prospective study of 681 patients with chronic LBP comparing chiropractic care to medical treatment with 18 months of follow-up. Although less than 20% of the patients experienced overall relief of pain and differences in outcome measures were not clinically significant, patients in the chiropractic group were more likely to perceive that their symptoms had improved.

Andersson et al.[270] compared osteopathic manual therapy to usual care in patients with a 3- to 6-month history of back pain and found that the overall results were similar, but the manual therapy group required fewer medications.

Other alternative medical therapies such as acupuncture, prolotherapy, and massage have also been evaluated. A Cochrane Review of acupuncture[271] showed superiority to placebo sham therapy and also a short-term benefit that did not extend beyond first follow-up when acupuncture was used in conjunction with other conventional therapies. However, a more recent systematic review by Ammendolia et al.[272] questioned inconclusive evidence of the success of acupuncture versus sham acupuncture and called for further randomized trials to rule out the possibility of a placebo effect.

Prolotherapy is a technique that attempts to regenerate ligamentous and tendinous structures of the spine via injections of various irritant solutions. The treatment is usually performed in conjunction with spinal manipulation. There is no consensus on method, type of solution injected, or frequency of sessions. Most practitioners use various combinations of saline, dextrose, glycerin, phenol, and lidocaine. Reports of the efficacy of prolotherapy are conflicted among many randomized trials and systematic reviews.[273-275] No evidence has been reported for the efficacy of prolotherapy without cointerventions such as spinal manipulation or exercise.

The efficacy of complementary and alternative modalities for the treatment of chronic LBP remains in question. The benefit of spinal manipulative therapy is also controversial but may improve patient satisfaction with care and perception of symptoms.

Pharmacotherapy

Judicious use of narcotic pain medications, oral steroids, and NSAIDs in patients with severe, acute back pain can provide good pain relief. The majority of patients with painful degenerative discs can be treated adequately on an outpatient basis. NSAIDs and acetaminophen are common over-the-counter medications used to treat back pain. A Cochrane Review[276] included 65 studies on NSAID use in LBP. NSAIDs were found to be superior to placebo but had significantly more side effects. There is no documented difference between type of NSAID, including cyclooxygenase-2 inhibitors. Acetaminophen has an effect similar to NSAIDs, but it has less risk of associated side effects when taken as directed and generally should be tried before NSAIDs. A Cochrane group concluded that NSAIDs are effective for short-term treatment of acute and chronic LBP, but the size of the effect is small.

Opioid formulations are also commonly used to treat back pain, but considering their widespread use there is a surprising paucity of high-quality randomized controlled data available on their efficacy. A Cochrane meta-analysis[277] of opioid use found only four studies, three of which focused on the use of tramadol. Pooled data found that tramadol, an atypical opioid, was more effective than placebo for pain relief and showed a slight improvement in functional scores. The only randomized controlled study of classic opioids[278] was a comparison to naproxen. Opioids were found to be more effective for pain relief but were not more effective for improving function than naproxen. The Cochrane authors concluded that the benefits of opioids for the treatment of chronic LBP are questionable, and further well-designed randomized controlled studies need to be performed. Two systematic reviews[279,280] of opioid use in the setting of chronic LBP have concluded there is evidence to support the efficacy of opioids for short-term relief of pain only. There is little evidence for long-term opioid use, which is fraught with an incidence of aberrant consumptive behavior approaching 25%.

Opioid pain medication use has many problems, ranging from minor side effects such as constipation and nausea to severe complications including respiratory depression, altered mental status, and insidious issues with tolerance and addiction. Another concern is the combination of opioids and acetaminophen in commonly prescribed formulations.[281] Hepatotoxicity and death from acetaminophen overdose is rare when the dose is less than 7.5 to 10 g in an 8-hour period,[282] but concern arises when patients inadvertently take larger doses in the setting of prescription drug abuse. An FDA advisory committee[283] recommended the addition of a boxed warning on the risk of acetaminophen overdose and also suggested elimination of combination opioid-acetaminophen formulations. Opioid pain medications should be given for only a few days in the setting of severe acute back pain. Opioid prescriptions for patients with chronic back pain are not recommended.

Oral tapering courses of steroids have been found to decrease symptoms of LBP in patients with radiculopathy from disc herniations.[175,284] In patients with axial LBP, steroids are not recommended.[285] Steroids can cause gastrointestinal bleeding; simultaneous use of gastrointestinal protective agents with oral steroids can reduce the risk of this complication.

Muscle relaxants are another class of medication routinely used in the treatment of muscle spasm associated with LBP. Their use should also be limited to very short courses due to their addictive potential. A Cochrane Review[286] of muscle relaxants for the treatment of back pain included 30 trials evaluating the use of benzodiazepines, nonbenzodiazepines, and antispasmotic muscle relaxants. Strong evidence for the efficacy of muscle relaxants over placebo was reported for short-term pain relief in the setting of acute back pain. No difference between the various drugs and classes was discerned. More trials to determine the efficacy of muscle relaxants compared with other analgesics and NSAIDs were recommended.

The last class of medications commonly prescribed in the setting of back pain are antidepressants. Their use may be particularly beneficial in patients presenting with chronic LBP in association with altered mental status, depression, anhedonia, sleep disturbances, agitation, and anorexia. Clinical studies[287-289] supporting the use of tricyclic antidepressants (TCAs) have shown an improvement in mood and sleep patterns. Low doses of TCAs also affect membrane potentials of peripheral nerves, which may be a mechanism by which they produce pain reduction. A 2003 review[290] of antidepressants in the treatment of chronic LBP found TCAs have a moderate effect on pain reduction in patients with no history of depression but reported conflicting evidence for improvement in functional outcomes. Physicians prescribing TCAs should be aware of potentially serious side effects involving orthostatic hypotension and cardiovascular perturbations. Selective serotonin reuptake inhibitors, another common class of antidepressants, failed to show efficacy in the treatment of chronic LBP and should be reserved for emotional or psychiatric disturbances related to back pain, not as a primary treatment for symptoms of back pain.[291]

Keller et al.[292] published a recent meta-analysis of nonsurgical management options for LBP. The authors reported that behavioral therapy, exercise therapy, and NSAIDs had the largest effect of those modalities studied. Machado et al.[293] published a separate large meta-analysis of placebo-controlled randomized trials of various forms of nonoperative treatment for nonspecific LBP. Small improvements in pain were found in patients treated with traction, physical therapy, antidepressants, and NSAIDs; moderate improvements were reported in patients treated with opioid analgesics, muscle relaxants, facet injections, and nerve blocks.

Nonsurgical Interventional Therapies

Nonsurgical treatment alternatives range from short-term temporizing measures, such as epidural injections, to procedures designed to be definitive treatments, such as intradiscal electrothermal therapy (IDET).

Epidural Spinal Injection

The advantage of epidural injections over oral steroids is the ability to achieve higher concentrations of steroid at the suspected site of pain while minimizing systemic effects. Epidural steroids typically work well when administered in the setting of radicular pain and not well in the setting of axial pain. Patients with foraminal stenosis secondary to loss of disc height may benefit from selective nerve root blocks either as a diagnostic or a therapeutic tool. Many clinicians recommend epidural steroid injections as second-line therapy in the treatment of lumbar disc disorders. Epidural steroids are commonly administered by three different routes: caudal, interlaminar, and transforaminal. The transforaminal approach is generally considered best because it achieves a better anterior epidural distribution. Complications from injection exist but are uncommon.[294,295]

Reports on the efficacy of epidural injections in the literature are contradictory. Manchikanti et al.[296] published preliminary results of a randomized trial of serial caudal epidural injections in patients with discogenic pain without disc herniation or radiculitis. The authors reported greater than 50% pain relief in 72% to 81% of patients and 40% reduction in ODI scores in 81% of patients. They concluded that caudal epidural injections with or without steroid are effective in treating discogenic back pain in more than 70% of patients. Two other observational studies[297,298] by the same authors had similar findings for the beneficial effects of caudal epidural injections in the specific setting of discogenic LBP.

Buttermann[299] studied patients with DDD and back pain of greater than 1 year duration who were candidates for fusion. There was initial success of treatment in greater than 50% of patients, but the success rate declined to between 23% and 29% by 1- to 2-year follow-up. The study had a high dropout rate, with more than two-thirds of the patients seeking another invasive treatment within 2 years. The authors concluded that patients with DDD but without spinal stenosis may experience a short-term benefit from epidural injections, with only one-fourth to one-third experiencing long-term improvement in pain and function. Other earlier studies of both caudal and transforaminal approaches have reported similar good short-term results, with up to 59% of patients having greater than 50% improvement in symptoms and function at a 1-year interval.[297,300]

A systematic review[301] criticized the literature on epidural injections for a lack of careful control of route of administration and patient diagnosis. In one evaluation of the pooled data, the only evidence found to support epidural injections was for short-term symptom relief in nonspecific LBP. No well-designed randomized trials were found specific to discogenic back pain. A 2008 Cochrane Review[302] of injection therapy for LBP failed to find sufficient evidence to make a recommendation. A systematic review by Chou et al.,[303] as part of the American Pain Society practice recommendations, found fair evidence that epidural steroid injection is moderately effective for short-term pain relief; however, the literature supporting its use in nonradicular LBP is sparse and has not shown significant benefit. No specific recommendation for the use of epidural steroid injections or the route of administration was made by the pain society.

Intradiscal Injection

Direct intradiscal injection, usually with a steroid solution, is another intervention that has been described in the literature for IDD. The desired effect is suppression of an inflammatory process within the disc, which is thought to be the cause of the discogenic pain. Intradiscal steroid injections were reported in an early case series by Feffer,[304] in which 47% of patients reportedly had remission of discogenic symptoms. Similar results were found by Wilkinson and Schuman.[305] Fayad et al.[306] reported short-term improvement in VAS score at a 1-month follow-up with intradiscal steroid injection in patients with Modic types 1 and 1 to 2 changes on MRI, but there was no long-term benefit. The only two major prospective randomized trials[307,308] of intradiscal steroid injection have failed to find a statistically significant benefit versus placebo in the treatment of discogenic back pain. Other authors have attempted intradiscal injection of various other substances, including solutions of chondroitin and dextrose,[309] hypertonic dextrose,[310] methylene blue,[311] and oxygen-ozone gas mixtures.[312,313] Although these studies report promising results, they have yet to be proven efficacious by rigorous randomized controlled trials.

Thermal Annuloplasty

Intradiscal electrothermal therapy (IDET) involves percutaneous insertion of a thermally controlled catheter into the IVD, usually the posterior anulus, and heating the tissue to a specific temperature (90°C) for a prescribed period of time (e.g., 20 minutes).

The proposed mechanism of action for these procedures is twofold: (1) elimination of nociceptive pain fibers and aberrant painful responses to the disrupted disc, and (2) collagen rearrangement in the anulus with resultant spinal segment stabilization. The biologic effects are not well understood and there is a lack of clear consensus regarding the effects on neuronal deafferentiation, collagen modulation, and spinal stability. Freeman et al.[314] studied the effect of nociceptor destruction via IDET on experimentally created anular tears in a sheep model. The authors failed to find any difference in the amount of neoinnervation in the anulus between specimens that underwent IDET and those that did not, which calls into question the theory of deafferentiation of the anulus. Whether collagen rearrangement with resultant shrinkage and stabilization of the discal element is a viable mechanism for IDET also remains in question.[315,316] Cadaveric studies of the effect of IDET on anular collagen have been performed by Kleinstueck et al.,[316] which showed a 10% to 16.7% reduction in tissue volume immediately adjacent to the electrode.

To destroy nociceptors in the anulus fibrosus temperatures must be raised to a minimum of 42°C to 45°C.[317,318] It is not possible to generate sufficient temperatures in the anulus with a radiofrequency probe placed in the center of the disc, as demonstrated by Houpt et al.[319] Temperature changes at distances further than 11 mm were insufficient to raise the tissue temperature of the outer anulus to the 42°C needed for neuronal ablation. Ashley et al.[320] compared temperature distribution in the disc between a radiofrequency needle and a navigable Spinecath. Using this method, they were able to deliver thermal energy to the anulus more effectively and achieved sufficient temperatures to cause denervation. Karasek and Bogduk[321] have recommended inserting the IDET electrode so as to remain within 5 mm of the outer surface of the anulus. Placement of the probe in the interlamellar plane rather than inside the innermost layer of the anulus allows for sufficient heat generation to destroy the nociceptors in the outer layers of the anulus.

Complications secondary to any of the thermal annulopasty procedures are rare. There has been one reported case of postoperative cauda equina syndrome caused by inadvertent placement of the catheter in the spinal canal,[322] and a few reports of broken catheters with no resultant adverse effect. There have been no reports of infection, bleeding, or other equipment- or technique-related complications.

Early uncontrolled clinical trials of IDET were promising, with improvement in 50% to 70% of patients,[137,321,323,324] but randomized controlled trials have produced conflicting results. Freeman et al.[325] found no significant improvement in outcome measures compared with sham surgery at 6-month follow-up. The opposite findings were reported by Pauza et al.[326] in patients with discographically diagnosed LBP of greater than 6 months' duration. The authors found that 40% of their patients who underwent IDET experienced at least 50% relief of pain, whereas a significant portion of the control group experienced symptom progression. They concluded that the IDET procedure is an effective intervention for a selective patient population and reported a needed-to-treat number of five to achieve 75% relief of pain. Barendse et al.[327] reported on a trial of intradiscal radiofrequency thermocoagulation in patients with chronic discogenic back pain. An 8-week follow-up assessment showed no difference from sham surgery in VAS score, global perceived effect, and ODI outcome measures.

Andersson et al.[328] published a systematic review of IDET versus spinal fusion in patients with disc degeneration and disruption. Similar median percentage improvement was noted between the two interventions for pain severity and quality-of-life outcomes. Fusion demonstrated better functional improvement but had a higher rate of complications. The authors concluded that IDET offers similar symptom relief with less risk of complications compared with fusion. Derby et al.,[329] in a systematic review, concluded that IDET is generally safer and cheaper than more invasive surgical techniques despite the fact that the best evidence available demonstrates only modest improvement in pain relief and functional outcomes.

Other systematic reviews of IDET have been more critical. Helm et al.[330] reported level 2 evidence in support of IDET in the setting of discogenic back pain based on two of the above randomized trials and a number of observational studies. Two observational studies were found by the authors in support of radiofrequency intradiscal thermocoagulation for level 3 evidence. Evidence in support of biacuplasty was lacking and was assigned as level 3 evidence. Freeman[331] published a systematic review of the literature that criticized generally poor outcomes even among studies in support of IDET. The author concluded that evidence for the efficacy of IDET is weak and has not passed the standard of scientific proof.

In summary of all nonoperative interventional therapies, Chou et al.[303] published a systematic review as part of the American Pain Society practice recommendations published in 2009. The authors reported fair evidence that epidural steroid injections are effective for short-term relief of pain. Good evidence was reported that prolotherapy, facet injection, intradiscal steroid, and intradiscal radiofrequency thermocoagulation are not effective. For IDET, no conclusions were made due to the fact that available randomized controlled trials are conflicting. IDET may best be indicated for less functionally impaired patients with well-maintained disc heights and discogenic pain from anular tears.[329]

Surgical Treatment

Once all conservative measures have been exhausted or if symptom nature warrants, surgical intervention may be required. The most common surgical treatment used for recalcitrant discogenic back pain and DDD is arthrodesis (fusion). Lumbar disc arthroplasty is another method approved by the FDA but is not currently in widespread use. Other motion-preserving options being investigated include dynamic stabilization of the motion segment and biologic disc repair.

Chou et al.[332] published a systematic review of surgical treatment for nonradicular LBP as part of the American Pain Society's practice recommendations. The authors found fair evidence that surgical fusion is no better than intensive rehabilitation with a cognitive behavioral emphasis. Surgically treated patients were noted as performing poorly, with less than 50% obtaining optimal outcome with fusion. The benefits of instrumented fusion compared with noninstrumented fusion were not clear. Fair evidence was found that arthroplasty performs as well as fusion for single-level DDD, but long-term outcome data were lacking.

The American Pain Society[223] practice recommendations, published in 2009, encourage clinicians to offer intensive interdisciplinary rehabilitation as an option with outcomes similar to surgery in the setting of nonradicular LBP. The report states that the majority of patients with nonradicular pain who undergo surgery do not experience an optimal outcome, which was defined by the pain society as (1) minimal or no pain, (2) discontinuation of or only occasional use of pain medications, and (3) return to high-level function. The society also suggested that there is insufficient evidence at this time to support disc arthroplasty for patients with nonradicular LBP. Other treatment guidelines also take a cautious view on spinal fusion for DDD, yet in some patients the symptoms are so severe that the chance for a good result makes surgical management attractive, especially when nonoperative treatment has failed.

Spinal Fusion

Surgical treatment for unremitting discogenic back pain has traditionally been spinal fusion. Most clinicians find that it is acceptable to perform spinal fusion for DDD in patients who have not responded to conservative care. The role of spinal fusion in the management of IDD is more controversial.[333]

Three high-quality randomized controlled studies have evaluated spinal fusion compared with nonoperative treatment in the setting of chronic LBP and DDD. Fritzell et al.[334] published a randomized controlled multicenter study of severe chronic LBP comparing fusion of the lower lumbar spine with nonsurgical therapy. The study involved a total of 222 operative and 72 nonoperative patients between the ages of 25 and 65 years with chronic LBP of at least 2 years' duration and radiologic evidence of disc degeneration at L4–L5, L5–S1, or both. The nonsurgical group received physical therapy, patient education, and alternative pain control modalities, such as TENS units, acupuncture, and injections. Results at 2-year follow-up were found to be significantly better in the fusion group, with back pain reduced by 33% compared with 7% in the nonsurgical group. Pain improvement was most significant during the first 6 months postoperatively but then gradually deteriorated thereafter. Disability according to ODI was reduced by 25% compared with 6% among nonsurgical patients, and 63% of surgical patients rated themselves as "much better" or "better" compared with 29% of nonsurgical patients. The "net back to work rate" was 36% in the surgical group and 13% in the nonsurgical group. The early complication rate in the surgical group was 17%. The authors concluded that surgical treatment of severe chronic LBP provides improved results compared with nonoperative treatment in carefully selected patients.

Brox et al.[251] published another randomized trial comparing outcomes of lumbar instrumented fusion versus cognitive intervention and exercise in 64 patients with chronic LBP and DDD. The critical component of this study was the addition of cognitive therapy to an intensive rehabilitation program. The mean change in ODI for the surgical fusion group was from 41 preoperatively to 26 at 1-year follow-up; mean change for the rehab group was from 42 to 30. The authors reported no significant difference in back pain, use of analgesics, emotional distress, and life satisfaction between the groups. Return-to-work rate at 1 year was 22% in the surgical group and 33% in the rehab group. The rehab group experienced greater improvement in fear-avoidance beliefs and fingertip-to-floor distance, while the surgical group had greater improvement in associated symptoms of leg pain. The overall success rate for surgical intervention was rated at 70% and 76% for nonoperative cognitive therapy. The authors concluded that there were near-equivalent outcomes between the groups, which was offset by an 18% complication rate among the surgical group.

The last major randomized clinical trial of surgery versus nonoperative therapy was published by Fairbank et al.[253] The MRC Spine Stabilization Trial was a randomized controlled trial comparing surgical treatment and intensive rehabilitation in 349 patients with chronic LBP. Similar to the Brox study, the intensive physical therapy program in the MRC trial also incorporated principles of cognitive behavioral therapy. At 1-year follow-up the mean ODI scores decreased from 46.5 to 34.0 in the surgical group and from 44.8 to 36.1 in the rehabilitation group. No significant differences were found between the groups in the shuttle walking test and Short Form-36 outcomes. The authors concluded that although the surgical group enjoyed a small—but statistically significant—benefit in one of the primary outcome measures (ODI), this was contradicted by the additional cost and potential risk of complication associated with surgery.

In a separate publication on the MRC trial, Rivero-Arias et al.[335] performed a cost analysis at 2-year follow-up. The cost per patient over the study time frame in the surgical group was estimated to be £7830 ($12,450), versus £4526 ($7200) in the rehabilitation group. There was no significant difference in mean quality-adjusted life-years between the groups. The authors concluded that surgical treatment was not a cost-effective use of health care funds compared with therapy, although the authors pointed out that ultimate costs could vary depending on the number of patients in either group that require subsequent intervention after the 2-year follow-up period.

Meta-analyses of surgical versus nonoperative treatment have paralleled the findings of Brox et al.[251] and the MRC trials.[253] Ibrahim et al.[336] pooled the data from these three randomized trials and found that a modest improvement in mean ODI scores among surgical patients should not be used as justification for routine operative treatment in light of a 16% early complication rate. Mirza et al.,[337] in a separate systematic review, concluded that surgical outcomes are equivalent to a structured rehabilitation program with cognitive behavioral therapy.

Phillips et al.[338] performed a systematic review of fusion in the treatment of DDD. They concluded that comparing surgery to nonoperative treatment obviates the fact that the treatments generally are performed in sequence, with surgery occurring when nonoperative treatment has failed. They further concluded that lumbar spine fusions result in meaningful improvements in pain and function.

Fusion Technologies

Over the years there has been a change from posterolateral and anterior stand-alone fusions to circumferential fusions. In recent years lateral fusion techniques have grown in popularity, as have minimally invasive fusion techiques. There is little question that minimally invasive approaches will grow in use; in fact, all procedures today are done with less-invasive techniques than in the past.

The use of new materials in implants and osteobiologics are discussed elsewhere in this text. Iliac crest bone grafts are today rarely used in the United States.

Adjacent-Segment Degeneration

A serious adverse effect of fusion surgery is adjacent-segment pathology. Sometimes it is an imaging finding without clinical importance, but sometimes it results in pain and may result in an extension of the fusion.[339,340]

As a consequence motion-preservation techniques have developed such as disc replacements and posterior motion preservation systems. These techniques are discussed in other chapters of this text.

Biologic Treatment of Degenerative Disc Disease

Over the past few decades research into biologic repair of disc degeneration has grown substantially. Molecular therapy, gene therapy, and stem cell disc regeneration are all viable alternatives but are not yet at the stage where they can be used clinically with predictable success.[341-343] Tissue engineering approaches are challenged by the complexity and stresses of the IVD.[344] Biologic repair will undoubtedly succeed at some time in the future but is not an immediate solution.

Summary

Lumbar disc disease is a common problem that affects many people at various ages in the form of IDD and DDD. Detailed history and physical examination are vital components in harmony with imaging modalities to make an accurate diagnosis. Recent practice guidelines have reaffirmed that the first line of treatment for those who have LBP of a discogenic source, with or without radicular symptoms, is conservative therapy. New emphasis has been placed on multidisciplinary therapy incorporating cognitive and behavioral treatment. Intradiscal therapy remains controversial, and many patients who undergo this procedure may eventually require arthrodesis. Surgical fusion, in all the various forms, is an appropriate option for patients who do not improve with appropriate nonoperative therapy. Preliminary studies of lumbar total disc replacement report equivalence to arthrodesis for the management of this patient population. Development of new motion-preserving techniques will likely change the treatment approach, as will emerging biologic techniques.

KEY POINTS

1. Detailed history and physical examination in conjunction with radiographic and MRI findings such as loss of disc height, disc signal changes, HIZs, and Modic changes are the best means available for diagnosing IDD and DDD.
2. Routine use of lumbar discography is not recommended for diagnosing a discogenic LBP.
3. Recent practice guidelines have reaffirmed that the first line of treatment for patients with LBP of a discogenic source, with or without radicular symptoms, is conservative therapy, and new emphasis has been placed on multidisciplinary therapy incorporating cognitive and behavioral treatment.
4. IDET remains controversial, and many patients who undergo this procedure still require arthrodesis. Other forms of less-invasive therapy, such as prolotherapy and intradiscal corticosteroid injections, are not recommended in this patient population.
5. Surgical fusion is an appropriate option for patients who do not improve with exhaustive nonoperative therapy.
6. No one method for achieving segmental fusion has clearly been shown to be better than another.
7. Preliminary studies of lumbar total disc replacement report equivalence to arthrodesis for the surgical management of this patient population, but there is still insufficient evidence to evaluate the long-term benefits and complications.

KEY REFERENCES

1. Aprill C, Bogduk N. High-intensity zone: a diagnostic sign of painful lumbar disc on magnetic resonance imaging. *Br J Radiol*. 1992;65(773):361-369.
   This study focuses on HIZ as a diagnostic sign of an anular tear.

2. Boden S, McCowin P, Davis D, et al. Abnormal magnetic-resonance scans of the lumbar spine in asymptomatic subjects: a prospective investigation. *J Bone Joint Surg Am*. 1990;72(3):403-408.
Abnormal MRI of the lumbar spine often reveals abnormalities in asymptomatic subjects, as illustrated by this article.
3. Brox J. Randomized clinical trial of lumbar instrumented fusion and cognitive intervention and exercises in patients with chronic low back pain and disc degeneration. *Spine*. 2003;28:1913-1921.
This article reports the outcomes of a randomized controlled trial of operative versus cognitive therapy and intensive exercise for the treatment of chronic low back pain and shows near-equivalent outcomes between the groups.
4. Carragee E. A gold standard evaluation of the "discogenic pain" diagnosis as determined by provocative discography. *Spine*. 2006;31:2115-2123.
This article calculates a best possible positive predictive value of lumbar discography in diagnosing discogenic back pain based on success of fusion in patients with positive discography versus a cohort of spondylolisthesis patients treated surgically.
5. Carragee E, Paragioudakis S, Khurana S. Lumbar high-intensity zone and discography in subjects without low back problems. *Spine*. 2000;25(23):2987-2992.
This article points out the low predictive value of lumbar HIZ and discography.
6. Chou R, Baisden J, Carragee EJ, et al. Surgery for low back pain: a review of the evidence for an American Pain Society Clinical Practice Guideline. *Spine*. 2009;34(10):1094-1109.
This meta-analysis reviews the best available evidence for surgical treatment of patients with low back pain as part of the American Pain Society's current practice guidelines.
7. Chou R, Loeser JD, Owens DK, et al. American Pain Society Low Back Pain Guideline Panel. Interventional therapies, surgery, and interdisciplinary rehabilitation for low back pain: an evidence-based clinical practice guideline from the American Pain Society. *Spine*. 2009;34(10):1066-1077.
This article reports the American Pain Society's current practice recommendations for the treatment of chronic low back pain.
8. Crock H. Internal disc disruption: a challenge to disc prolapse fifty years on. *Spine*. 1986;11(6):650-653.
The concept of IDD was elaborated in this article.
9. Fairbank J. Randomised controlled trial to compare surgical stabilisation of the lumbar spine with an intensive rehabilitation programme for patients with chronic low back pain: the MRC spine stabilisation trial. *BMJ*. 2005;330:1233.
This large randomized controlled trial compared operative and interdisciplinary physical therapy programs in the treatment of chronic low back pain. A small benefit was shown for the surgically treated group but with a significant increase in cost and complications.
10. Fritzell P, et al. Lumbar fusion versus nonsurgical treatment for chronic low back pain: a multicenter randomized controlled trial from the Swedish Lumbar Spine Study Group. *Spine*. 2001;26(23):2521-2532.
This multicenter, randomized controlled trial comparing lumbar fusion and nonsurgical treatment for chronic back pain showed superiority for the fusion alternative.
11. Modic M, Steinberg P, Ross J, et al. Degenerative disc disease: assessment of changes in vertebral body marrow with MR imaging. *Radiology*. 1988;166(1 Pt 1):193-199.
The changes observed in vertebrae adjacent to the degenerative disc are described and classified.

REFERENCES

1. Arnbak B, Jensen TS, Egund N, et al. Prevalence of degenerative and spondyloarthritis-related magnetic resonance imaging findings in the spine and sacroiliac joints in patients with persistent low back pain. *Eur Radiol*. 2016;26(4):1191-1203.
2. Livshits G, Popham M, Malkin I, et al. Lumbar disc degeneration and genetic factors are the main risk factors for low back pain in women: the UK Twin Spine Study. *Ann Rheum Dis*. 2011;70(10):1740-1745. doi:10.1136/ard.2010.137836.
3. Cheung KM, Karppinen J, Chan D, et al. Prevalence and pattern of lumbar magnetic resonance imaging changes in a population study of one thousand forty-three individuals. *Spine*. 2009;34(9):934-940.
4. Dreischarf M, Albiol L, Rohlmann A, et al. Age-related loss of lumbar spinal lordosis and mobility–a study of 323 asymptomatic volunteers. *PLoS One*. 2014;9(12):e116186.
5. Andersson G. Epidemiological features of chronic low-back pain. *Lancet*. 1999;354:581-585.
6. Papageorgiou AC, Croft PR, Ferry S, Jayson MI, Silman AJ. Estimating the prevalence of low back pain in the general population. Evidence from the South Manchester Back Pain Survey. *Spine*. 1995;20:1889-1894.
7. Waddell G. A new clinical model for the treatment of low-back pain. *Spine*. 1987;12:632-644.
8. Croft PR, Macfarlane GJ, Papageorgiou AC, Thomas E, Silman AJ. Outcome of low back pain in general practice: a prospective study. *BMJ*. 1998;316:1356-1359.
9. Walsh K, Cruddas M, Coggon D. Low back pain in eight areas of Britain. *J Epidemiol Community Health*. 1992;46:227-230.
10. Andersson GB, Svensson HO, Odén A. The intensity of work recovery in low back pain. *Spine*. 1983;8:880-884.
11. Wahlgren DR, Atkinson JH, Epping-Jordan JE, et al. One-year follow-up of first onset low back pain. *Pain*. 1997;73:213-221.
12. Carey TS, Garrett JM, Jackman AM. Beyond the good prognosis. Examination of an inception cohort of patients with chronic low back pain. *Spine*. 2000;25:115-120.
13. Roland MO, Morrell DC, Morris RW. Can general practitioners predict the outcome of episodes of back pain? *Br Med J (Clin Res Ed)*. 1983;286:523-525.
14. Papageorgiou AC, Croft PR, Thomas E, et al. Influence of previous pain experience on the episode incidence of low back pain: results from the South Manchester Back Pain Study. *Pain*. 1996;66:181-185.
15. Smith SE, Darden BV, Rhyne AL, Wood KE. Outcome of unoperated discogram-positive low back pain. *Spine*. 1995;20:1997-2000.
16. Kirkaldy-Willis WH, Farfan HF. Instability of the lumbar spine. *Clin Orthop Relat Res*. 1982;165:110-123.
17. Waris E, et al. Disc degeneration in low back pain: a 17-year follow-up study using magnetic resonance imaging. *Spine*. 2007;32:681-684.
18. Dahia CL, Mahoney EJ, Durrani AA, Wylie C. Intercellular signaling pathways active during intervertebral disc growth, differentiation, and aging. *Spine*. 2009;34:456-462.
19. Risbud MV, Shapiro IM. Notochordal cells in the adult intervertebral disc: new perspective on an old question. *Crit Rev Eukaryot Gene Expr*. 2011;21(1):29-41.
20. Gan JC, Ducheyne P, Vresilovic EJ, Swaim W, Shapiro IM. Intervertebral disc tissue engineering I: characterization of the nucleus pulposus. *Clin Orthop Relat Res*. 2003;411:305-314.

21. Risbud MV, Schipani E, Shapiro IM. Hypoxic regulation of nucleus pulposus cell survival: from niche to notch. *Am J Pathol*. 2010;176(4):1577-1583.
22. Agrawal A, Guttapalli A, Narayan S, et al. Normoxic stabilization of HIF-1alpha drives glycolytic metabolism and regulates aggrecan gene expression in nucleus pulposus cells of the rat intervertebral disk. *Am J Physiol Cell Physiol*. 2007;293:C621-C631.
23. Souter WA, Taylor TK. Sulphated acid mucopolysaccharide metabolism in the rabbit intervertebral disc. *J Bone Joint Surg Br*. 1970;52:371-384.
24. Marchand F, Ahmed AM. Investigation of the laminate structure of lumbar disc anulus fibrosus. *Spine*. 1990;15:402-410.
25. Arana CJ, Diamandis EP, Kandel RA. Cartilage tissue enhances proteoglycan retention by nucleus pulposus cells in vitro. *Arthritis Rheum*. 2010;62(11):3395-3403.
26. Kim KW, Lim TH, Kim JG, et al. The origin of chondrocytes in the nucleus pulposus and histologic findings associated with the transition of a notochordal nucleus pulposus to a fibrocartilaginous nucleus pulposus in intact rabbit intervertebral discs. *Spine*. 2003;28(10):982-990.
27. Kim KW, Ha KY, Park JB, et al. Expressions of membrane-type I matrix metalloproteinase, Ki-67 protein, and type II collagen by chondrocytes migrating from cartilage endplate into nucleus pulposus in rat intervertebral discs: a cartilage endplate-fracture model using an intervertebral disc organ culture. *Spine*. 2005;30(12):1373-1378.
28. Moore RJ. The vertebral end-plate: what do we know? *Eur Spine J*. 2000;9:92-96.
29. Nachemson A, Lewin T, Maroudas A, Freeman MA. In vitro diffusion of dye through the end-plates and the annulus fibrosus of human lumbar inter-vertebral discs. *Acta Orthop Scand*. 1970;41:589-607.
30. Crock HV, Yoshizawa H. The blood supply of the lumbar vertebral column. *Clin Orthop Relat Res*. 1976;115:6-21.
31. Gruber HE, Ashraf N, Kilburn J, et al. Vertebral endplate architecture and vascularization: application of micro-computerized tomography, a vascular tracer, and immunocytochemistry in analyses of disc degeneration in the aging sand rat. *Spine*. 2005;30:2593-2600.
32. Hassler O. The human intervertebral disc: A micro-angiographical study on its vascular supply at various ages. *Acta Orthop Scand*. 1969;40:765-772.
33. Rudert M, Tillmann B. Lymph and blood supply of the human intervertebral disc: cadaver study of correlations to discitis. *Acta Orthop Scand*. 1993;64:37-40.
34. Ejeskär A, Holm S. Oxygen tension measurements in the intervertebral disc. A methodological and experimental study. *Ups J Med Sci*. 1979;84(1):83-93.
35. Bartels EM, Fairbank JC, Winlove CP, Urban JP. Oxygen and lactate concentrations measured in vivo in the intervertebral discs of patients with scoliosis and back pain. *Spine*. 1998; 23:1-7.
36. Rajpurohit R, Risbud MV, Ducheyne P, Vresilovic EJ, Shapiro IM. Phenotypic characteristics of the nucleus pulposus: expression of hypoxia inducing factor-1, glucose transporter-1 and MMP-2. *Cell Tissue Res*. 2002;308:401-407.
37. Semenza GL, Roth PH, Fang HM, Wang GL. Transcriptional regulation of genes encoding glycolytic enzymes by hypoxia-inducible factor 1. *J Biol Chem*. 1994;269:23757-23763.
38. Wang GL, Jiang BH, Rue EA, Semenza GL. Hypoxia-inducible factor 1 is a basic-helix-loop-helix-PAS heterodimer regulated by cellular O2 tension. *Proc Natl Acad Sci USA*. 1995;92:5510-5514.
39. Papandreou I, Cairns RA, Fontana L, Lim AL, Denko NC. HIF-1 mediates adaptation to hypoxia by actively downregulating mitochondrial oxygen consumption. *Cell Metab*. 2006;3:187-197.
40. Fukuda R, Zhang H, Kim JW, et al. HIF-1 regulates cytochrome oxidase subunits to optimize efficiency of respiration in hypoxic cells. *Cell*. 2007;129:111-122.
41. Schipani E, Ryan HE, Didrickson S, et al. Hypoxia in cartilage: HIF-1 alpha is essential for chondrocyte growth arrest and survival. *Genes Dev*. 2001;15:2865-2876.
42. Zhang H, Bosch-Marce M, Shimoda LA, et al. Mitochondrial autophagy is an HIF-1–dependent adaptive metabolic response to hypoxia. *J Biol Chem*. 2008;283:10892-10903.
43. Hofbauer KH, Gess B, Lohaus C, et al. Oxygen tension regulates the expression of a group of procollagen hydroxylases. *Eur J Biochem*. 2003;270:4515-4522.
44. Sowter HM, Raval RR, Moore JW, Ratcliffe PJ, Harris AL. Predominant role of hypoxia-inducible transcription factor (Hif)-1 alpha versus Hif-2 alpha in regulation of the transcriptional response to hypoxia. *Cancer Res*. 2003;63:6130-6134.
45. Smits P, Lefebvre V. Sox5 and Sox6 are required for notochord extracellular matrix sheath formation, notochord cell survival and development of the nucleus pulposus of intervertebral discs. *Development*. 2003;130:1135-1148.
46. Lafont JE, Talma S, Murphy CL. Hypoxia-inducible factor 2alpha is essential for hypoxic induction of the human articular chondrocyte phenotype. *Arthritis Rheum*. 2007;56:3297-3306.
47. Khan WS, Adesida AB, Hardingham TE. Hypoxic conditions increase hypoxia-inducible transcription factor 2alpha and enhance chondrogenesis in stem cells from the infrapatellar fat pad of osteoarthritis patients. *Arthritis Res Ther*. 2007;9:R55.
48. Kanichai M, Ferguson D, Prendergast PJ, Campbell VA. Hypoxia promotes chondrogenesis in rat mesenchymal stem cells: a role for AKT and hypoxia-inducible factor (HIF)-1alpha. *J Cell Physiol*. 2008;216:708-715.
49. Risbud MV, Guttapalli A, Stokes DG, et al. Nucleus pulposus cells express HIF-1alpha under normoxic culture conditions: a metabolic adaptation to the intervertebral disc microenvironment. *J Cell Biochem*. 2006;98:152-159.
50. Agrawal A, Gajghate S, Smith H, et al. Cited2 modulates hypoxia-inducible factor-dependent expression of vascular endothelial growth factor in nucleus pulposus cells of the rat intervertebral disc. *Arthritis Rheum*. 2008;58:3798-3808.
51. Merceron C, Mangiavini L, Robling A, et al. Loss of HIF-1α in the notochord results in cell death and complete disappearance of the nucleus pulposus. *PLoS One*. 2014;9:e110768.
52. Risbud MV, Shapiro IM. Role of cytokines in intervertebral disc degeneration: pain and disc content. *Nat Rev Rheumatol*. 2014;10:44-56.
53. Roberts S, Urban JPG. Intervertebral discs. In: Riihimäki H, Viikari-Juntura E, eds. *Encyclopedia of Occupational Health and Safety*. Geneva: International Labor Organization; 2011.
54. Ishihara H, Warensjo K, Roberts S, Urban JP. Proteoglycan synthesis in the intervertebral disk nucleus: the role of extracellular osmolality. *Am J Physiol*. 1997;272:C1499-C1506.
55. Van Dijk B, Potier E, Ito K. Culturing bovine nucleus pulposus explants by balancing medium osmolarity. *Tissue Eng C Methods*. 2011;17:1089-1096.

56. Wuertz K, Urban JPG, Klasen J, et al. Influence of extracellular osmolarity and mechanical stimulation on gene expression of intervertebral disc cells. *J Orthop Res*. 2007;25:1513-1522.
57. Neidlinger-Wilke C, Mietsch A, Rinkler C, et al. Interactions of environmental conditions and mechanical loads have influence on matrix turnover by nucleus pulposus cells. *J Orthop Res*. 2012;30:112-121.
58. Haschtmann D, Stoyanov JV, Ferguson SJ. Influence of diurnal hyperosmotic loading on the metabolism and matrix gene expression of a whole-organ intervertebral disc model. *J Orthop Res*. 2006;24:1957-1966.
59. Urban JP. The chondrocyte: a cell under pressure. *Br J Rheumatol*. 1994;33:901-908.
60. Roberts N, Hogg D, Whitehouse GH, Dangerfield P. Quantitative analysis of diurnal variation in volume and water content of lumbar intervertebral discs. *Clin Anat*. 1998;11:1-8.
61. Tsai TT, Danielson KG, Guttapalli A, et al. TonEBP/OREBP is a regulator of nucleus pulposus cell function and survival in the intervertebral disc. *J Biol Chem*. 2006;281:25416-25424.
62. Hiyama A, Gogate SS, Gajghate S, et al. BMP-2 and TGF-beta stimulate expression of beta1,3-glucuronosyl transferase 1 (GlcAT-1) in nucleus pulposus cells through AP1, TonEBP, and Sp1: role of MAPKs. *J Bone Miner Res*. 2010;25:1179-1190.
63. Gajghate S, Hiyama A, Shah M, et al. Osmolarity and intracellular calcium regulate aquaporin2 expression through TonEBP in nucleus pulposus cells of the intervertebral disc. *J Bone Miner Res*. 2009;24:992-1001.
64. Johnson ZI, Shapiro IM, Risbud MV. Extracellular osmolarity regulates matrix homeostasis in the intervertebral disc and articular cartilage: evolving role of TonEBP. *Matrix Biol*. 2014;40:10-16.
65. Nakamura SI, et al. The afferent pathways of discogenic low-back pain. Evaluation of L2 spinal nerve infiltration. *J Bone Joint Surg Br*. 1996;78:606-612.
66. Buckwalter JA. Aging and degeneration of the human intervertebral disc. *Spine*. 1995;20:1307-1314.
67. Crock HV. Internal disc disruption. A challenge to disc prolapse fifty years on. *Spine*. 1986;11:650-653.
68. Macnab I. The traction spur. An indicator of segmental instability. *J Bone Joint Surg Am*. 1971;53:663-670.
69. Hangai M, Kaneoka K, Kuno S, et al. Factors associated with lumbar intervertebral disc degeneration in the elderly. *Spine J*. 2008;8:732-740.
70. Kalichman L, Hunter DJ. The genetics of intervertebral disc degeneration. Familial predisposition and heritability estimation. *Joint Bone Spine*. 2008;75:383-387.
71. Battié MC, Videman T, Gibbons LE, et al. 1995 Volvo Award in clinical sciences. Determinants of lumbar disc degeneration. A study relating lifetime exposures and magnetic resonance imaging findings in identical twins. *Spine*. 1995;20:2601-2612.
72. An HS, Silveri CP, Simpson JM, et al. Comparison of smoking habits between patients with surgically confirmed herniated lumbar and cervical disc disease and controls. *J Spinal Disord*. 1994;7:369-373.
73. Kauppila LI. Atherosclerosis and disc degeneration/low-back pain—a systematic review. *Eur J Vasc Endovasc Surg*. 2009;37:661-670.
74. Hassett G, Hart DJ, Manek NJ, Doyle DV, Spector TD. Risk factors for progression of lumbar spine disc degeneration: the Chingford Study. *Arthritis Rheum*. 2003;48(11):3112-3117.
75. Takatalo J, Karppinen J, Taimela S, et al. Body mass index is associated with lumbar disc degeneration in young Finnish males: subsample of Northern Finland birth cohort study 1986. *BMC Musculoskelet Disord*. 2013;14:87.
76. Lis AM, Black KM, Korn H, Nordin M. Association between sitting and occupational LBP. *Eur Spine J*. 2007;16:283-298.
77. Kelsey JL, Hardy RJ. Driving of motor vehicles as a risk factor for acute herniated lumbar intervertebral disc. *Am J Epidemiol*. 1975;102:63-73.
78. Wilder DG. The biomechanics of vibration and low back pain. *Am J Ind Med*. 1993;23:577-588.
79. Arun R, Freeman BJ, Scammell BE, et al. What influence does sustained mechanical load have on diffusion in the human intervertebral disc? an in vivo study using serial postcontrast magnetic resonance imaging. *Spine*. 2009;34:2324-2337.
80. Videman T, Battié MC, Ripatti S, et al. Determinants of the progression in lumbar degeneration: a 5-year follow-up study of adult male monozygotic twins. *Spine*. 2006;31:671-678.
81. MacGregor AJ, Andrew T, Sambrook PN, Spector TD. Structural, psychological, and genetic influences on low back and neck pain: a study of adult female twins. *Arthritis Rheum*. 2004;51:160-167.
82. Hestbaek L, Iachine IA, Leboeuf-Yde C, Kyvik KO, Manniche C. Heredity of low back pain in a young population: a classical twin study. *Twin Res*. 2004;7:16-26.
83. Hartvigsen J, Christensen K, Frederiksen H, Petersen HC. Genetic and environmental contributions to back pain in old age: a study of 2,108 Danish twins aged 70 and older. *Spine*. 2004;29:897-901.
84. Paassilta P, Lohiniva J, Göring HH, et al. Identification of a novel common genetic risk factor for lumbar disc disease. *JAMA*. 2001;285:1843-1849.
85. Jim JJ, Noponen-Hietala N, Cheung KM, et al. The TRP2 allele of COL9A2 is an age-dependent risk factor for the development and severity of intervertebral disc degeneration. *Spine*. 2005;30:2735-2742.
86. Solovieva S, Lohiniva J, Leino-Arjas P, et al. Intervertebral disc degeneration in relation to the COL9A3 and the IL-1ss gene polymorphisms. *Eur Spine J*. 2006;15:613-619.
87. Kalichman L, Hunter DJ. The genetics of intervertebral disc degeneration. Associated genes. *Joint Bone Spine*. 2008;75:388-396.
88. Williams FM, Bansal AT, van Meurs JB, et al. Novel genetic variants associated with lumbar disc degeneration in northern Europeans: a meta-analysis of 4600 subjects. *Ann Rheum Dis*. 2013;72(7):1141-1148.
89. Song YQ, Karasugi T, Cheung KM, et al. Lumbar disc degeneration is linked to a carbohydrate sulfotransferase 3 variant. *J Clin Invest*. 2013;123(11):4909-4917.
90. Crock HV. A reappraisal of intervertebral disc lesions. *Med J Aust*. 1970;1:983-989.
91. Haughton VM. MR imaging of the spine. *Radiology*. 1988;166:297-301.
92. Cuellar JM, Stauff MP, Herzog RJ, et al. Does provocative discography cause clinically important injury to the lumbar intervertebral disc? A 10-year matched cohort study. *Spine J*. 2016;16(3):273-280.
93. Borthakur A, Maurer PM, Fenty M, et al. T1ρ magnetic resonance imaging and discography pressure as novel biomarkers for disc degeneration and low back pain. *Spine*. 2011;36(25):2190-2196.

94. Binch AL, Cole AA, Breakwell LM, et al. Nerves are more abundant than blood vessels in the degenerate human intervertebral disc. *Arthritis Res Ther*. 2015;17:370.
95. Maidhof R, Jacobsen T, Papatheodorou A, Chahine NO. Inflammation induces irreversible biophysical changes in isolated nucleus pulposus cells. *PLoS One*. 2014;9:1-11.
96. Ellman MB, Kim J-S, An HS, et al. Toll-like receptor adaptor signaling molecule MyD88 on intervertebral disk homeostasis: in vitro, ex vivo studies. *Gene*. 2012;505:283-290.
97. Le Maitre CL, Freemont AJ, Hoyland JA. The role of interleukin-1 in the pathogenesis of human intervertebral disc degeneration. *Arthritis Res Ther*. 2005;7:R732-R745.
98. Wang J, Markova D, Anderson DG, et al. TNF-α and IL-1β promote a disintegrin-like and metalloprotease with thrombospondin type I motif-5-mediated aggrecan degradation through syndecan-4 in intervertebral disc. *J Biol Chem*. 2011;286:39738-39749.
99. Wang X, Wang H, Yang H, et al. Tumor necrosis factor-α- and interleukin-1β-dependent matrix metalloproteinase-3 expression in nucleus pulposus cells requires cooperative signaling via syndecan 4 and mitogen-activated protein kinase-nuclear factor κB axis: implications in inflammatory disease. *Am J Pathol*. 2014;184:1-13.
100. Murata Y, Rydevik B, Nannmark U, et al. Local application of interleukin-6 to the dorsal root ganglion induces tumor necrosis factor-α in the dorsal root ganglion and results in apoptosis of the dorsal root ganglion cells. *Spine (Phila Pa)*. 2011;36:926-932.
101. Burke JG, Watson RWG, McCormack D, et al. Intervertebral discs which cause low back pain secrete high levels of proinflammatory mediators. *J Bone Joint Surg Br*. 2002;84:196-201.
102. Abe Y, Akeda K, An HS, et al. Proinflammatory cytokines stimulate the expression of nerve growth factor by human intervertebral disc cells. *Spine (Phila Pa)*. 2007;32:635-642.
103. García-Cosamalón J, del Valle ME, Calavia MG, et al. Intervertebral disc, sensory nerves and neurotrophins: who is who in discogenic pain? *J Anat*. 2010;217:1-15.
104. Purmessur D, Freemont AJ, Hoyland JA. Expression and regulation of neurotrophins in the nondegenerate and degenerate human intervertebral disc. *Arthritis Res Ther*. 2008;10:R99.
105. Kepler CK, Markova DZ, Hilibrand AS, et al. Substance P stimulates production of inflammatory cytokines in human disc cells. *Spine J*. 2012;12:S109.
106. Carragee EJ, Kim D, van der Vlugt T, Vittum D. The clinical use of erythrocyte sedimentation rate in pyogenic vertebral osteomyelitis. *Spine (Phila Pa)*. 1997;18:2089-2093.
107. Sapico FL, Montgomerie JZ. Pyogenic vertebral osteomyelitis: report of nine cases and review of the literature. *Rev Infect Dis*. 1979;1:754-776.
108. Modic MT, Steinberg PM, Ross JS, Masaryk TJ, Carter JR. Degenerative disk disease: assessment of changes in vertebral body marrow with MR imaging. *Radiology*. 1988;166:193-199.
109. Jensen TS, Karppinen J, Sorensen JS, Niinimäki J, Leboeuf-Yde C. Vertebral endplate signal changes (Modic change): A systematic literature review of prevalence and association with non-specific low back pain. *Eur Spine*. 2008;17:1407-1422.
110. Albert HB, Lambert P, Rollason J, et al. Does nuclear tissue infected with bacteria following disc herniations lead to Modic changes in the adjacent vertebrae? *Eur Spine J*. 2013;22:690-696.
111. Agarwal VJ, Golish R, Kondrashov D, Alamin TF. Results of bacterial culture from surgically excised intervertebral disc in 52 patients undergoing primary lumbar microdiscectomy at a single level. *Spine J*. 2010;10:S45-S46.
112. Fritzell P, Bergström T, Welinder-Olsson C. Detection of bacterial DNA in painful degenerated spinal discs in patients without signs of clinical infection. *Eur Spine J*. 2004;13:702-706.
113. Albert HB, Sorensen JS, Christensen BS, Manniche C. Antibiotic treatment in patients with chronic low back pain and vertebral bone edema (Modic type 1 changes): a double-blind randomized clinical controlled trial of efficacy. *Eur Spine J*. 2013;22:697-707.
114. Casellas F, Borruel N, Papo M, et al. Antiinflammatory effects of enterically coated amoxicillin-clavulanic acid in active ulcerative colitis. *Inflamm Bowel Dis*. 1998;4:1-5.
115. Hajhashemi V, Dehdashti K. Antinociceptive effect of clavulanic acid and its preventive activity against development of morphine tolerance and dependence in animal models. *Res Pharm Sci*. 2014;9:315-321.
116. Roca I, Akova M, Baquero F, et al. The global threat of antimicrobial resistance: Science for intervention. *New Microbes New Infect*. 2015;6:22-29.
117. Quero L, Klawitter M, Schmaus A, et al. Hyaluronic acid fragments enhance the inflammatory and catabolic response in human intervertebral disc cells through modulation of toll-like receptor 2 signalling pathways. *Arthritis Res Ther*. 2013;15:R94.
118. Kawagoe T, Sato S, Matsushita K, et al. Sequential control of Toll-like receptor-dependent responses by IRAK1 and IRAK2. *Nat Immunol*. 2008;9:684-691.
119. Scheibner KA, Lutz MA, Boodoo S, et al. Hyaluronan fragments act as an endogenous danger signal by engaging TLR2. *J Immunol*. 2006;177:1272-1281.
120. Campo GM, Avenoso A, D'Ascola A, et al. Hyaluronan differently modulates TLR-4 and the inflammatory response in mouse chondrocytes. *Biofactors*. 2012;38:69-76.
121. Campo GM, Avenoso A, Campo S, et al. Small hyaluronan oligosaccharides induce inflammation by engaging both toll-like-4 and CD44 receptors in human chondrocytes. *Biochem Pharmacol*. 2010;80:480-490.
122. Scaffidi P, Misteli T, Bianchi ME. Release of chromatin protein HMGB1 by necrotic cells triggers inflammation. *Nature*. 2002;418:191-195.
123. Voll RE, Urbonaviciute V, Herrmann M, Kalden JR. High mobility group box 1 in the pathogenesis of inflammatory and autoimmune diseases. *Isr Med Assoc J*. 2008;10:26-28.
124. Tang D, Kang R, Cheh CW, et al. HMGB1 release and redox regulates autophagy and apoptosis in cancer cells. *Oncogene*. 2010;29(38):5299-5310.
125. Otoshi K, Kikuchi S, Kato K, Sekiguchi M, Konno S. Anti-HMGB1 neutralization antibody improves pain-related behavior induced by application of autologous nucleus pulposus onto nerve roots in rats. *Spine (Phila Pa)*. 2011;36:E692-E698.
126. Ruel N, Markova DZ, Adams SL, et al. Fibronectin fragments and the cleaving enzyme ADAM-8 in the degenerative human intervertebral disc. *Spine (Phila Pa)*. 2014;39:1274-1279.
127. Liu H-F, Zhang H, Qiao G-X, et al. A novel rabbit disc degeneration model induced by fibronectin fragment. *Joint Bone Spine*. 2013;80:301-306.

128. Anderson DG, Li X, Balian GA. Fibronectin fragment alters the metabolism by rabbit intervertebral disc cells in vitro. *Spine (Phila Pa)*. 2005;30:1242-1246.
129. Xia M, Zhu Y. Fibronectin fragment activation of ERK increasing integrin α5 and β1 subunit expression to degenerate nucleus pulposus cells. *J Orthop Res*. 2011;29:556-561.
130. Tran CM, Schoepflin ZR, Markova DZ, et al. CCN2 suppresses catabolic effects of interleukin-1β through α5β1 and αVβ3 integrins in nucleus pulposus cells: implications in intervertebral disc degeneration. *J Biol Chem*. 2014;289:7374-7387.
131. Fields AJ, Berg-Johansen B, Metz LN, et al. Alterations in intervertebral disc composition, matrix homeostasis and biomechanical behavior in the UCD-T2DM rat model of type 2 diabetes. *J Orthop Res*. 2015;33:738-746.
132. Tsai T-T, Ho NY-J, Lin Y-T, et al. Advanced glycation end products in degenerative nucleus pulposus with diabetes. *J Orthop Res*. 2014;32:238-244.
133. Illien-Junger S, Grosjean F, Laudier DM, et al. Combined anti-inflammatory and anti-AGE drug treatments have a protective effect on intervertebral discs in mice with diabetes. *PLoS One*. 2013;8:e64302.
134. Johnson WEB, Stephan S, Roberts S. The influence of serum, glucose and oxygen on intervertebral disc cell growth in vitro: implications for degenerative disc disease. *Arthritis Res Ther*. 2008;10:R46.
134a. Videman T, Battie MC, Gibbons LE, et al. Disc degeneration and bone density in monozygotic twins discordant for insulin-dependent diabetes mellitus. *J Orthop Res*. 2000;18(5):768-772.
134b. Fabiane SM, Ward KJ, Iatridis JC, et al. Does type 2 diabetes mellitus promote intervertebral disc degeneration? *Eur Spine J*. 2016;25(9):2716-2720.
135. Deyo RA, et al. Outcome measures for low back pain research. A proposal for standardized use. *Spine*. 1998;23:2003-2013.
136. Spruit M, Jacobs WC. Pain and function after intradiscal electrothermal treatment (IDET) for symptomatic lumbar disc degeneration. *Eur Spine J*. 2002;11:589-593.
137. Derby R, Eek B, Chen Y. Intradiscal electrothermal annuloplasty (IDET): a novel approach for treating chronic discogenic back pain. *Neuromodulation*. 2000;3(2):82-88.
138. Weinstein J, Claverie W, Gibson S. The pain of discography. *Spine*. 1988;13:1344-1348.
139. Weinstein J. Neurogenic and nonneurogenic pain and inflammatory mediators. *Orthop Clin North Am*. 1991;22:235-246.
140. Kawakami M, et al. Possible mechanism of painful radiculopathy in lumbar disc herniation. *Clin Orthop Relat Res*. 1998;351:241-251.
141. Byröd G, et al. A rapid transport route between the epidural space and the intraneural capillaries of the nerve roots. *Spine*. 1995;20:138-143.
142. Freemont AJ, et al. Nerve ingrowth into diseased intervertebral disc in chronic back pain. *Lancet*. 1997;350:178-181.
143. Evans W, Jobe W, Seibert C. A cross-sectional prevalence study of lumbar disc degeneration in a working population. *Spine*. 1989;14:60-64.
144. Gibson MJ, et al. Magnetic resonance imaging and discography in the diagnosis of disc degeneration. A comparative study of 50 discs. *J Bone Joint Surg Br*. 1986;68:369-373.
145. Saal JS, et al. High levels of inflammatory phospholipase A2 activity in lumbar disc herniations. *Spine*. 1990;15:674-678.
146. Nygaard OP, Mellgren SI, Osterud B. The inflammatory properties of contained and noncontained lumbar disc herniation. *Spine*. 1997;22:2484-2488.
147. O'Donnell JL, O'Donnell AL. Prostaglandin E2 content in herniated lumbar disc disease. *Spine*. 1996;21:1653-1655.
148. Franson RC, Saal JS, Saal JA. Human disc phospholipase A2 is inflammatory. *Spine*. 1992;17(suppl 6):S129-S132.
149. Burke JG, Watson RW, McCormack D, et al. Intervertebral discs which cause low back pain secrete high levels of proinflammatory mediators. *J Bone Joint Surg Br*. 2002;84:196-201.
150. Igarashi A, Kikuchi S, Konno S, Olmarker K. Inflammatory cytokines released from the facet joint tissue in degenerative lumbar spinal disorders. *Spine*. 2004;29:2091-2095.
151. Ohtori S, et al. Tumor necrosis factor-immunoreactive cells and PGP 9.5-immunoreactive nerve fibers in vertebral endplates of patients with discogenic low back pain and Modic type 1 or type 2 changes on MRI. *Spine*. 2006;31:1026-1031.
152. Ashton IK, et al. Substance P in intervertebral discs. Binding sites on vascular endothelium of the human annulus fibrosus. *Acta Orthop Scand*. 1994;65:635-639.
153. Palmgren T, et al. An immunohistochemical study of nerve structures in the anulus fibrosus of human normal lumbar intervertebral discs. *Spine*. 1999;24:2075-2079.
154. Coppes MH, et al. Innervation of "painful" lumbar discs. *Spine*. 1997;22:2342-2349.
155. Peng B, et al. The pathogenesis of discogenic low back pain. *J Bone Joint Surg Br*. 2005;87:62-67.
156. Brown MF, et al. Sensory and sympathetic innervation of the vertebral endplate in patients with degenerative disc disease. *J Bone Joint Surg Br*. 1997;79:147-153.
157. Takebayashi T, et al. Sympathetic afferent units from lumbar intervertebral discs. *J Bone Joint Surg Br*. 2006;88:554-557.
158. Aoki Y, et al. Innervation of the lumbar intervertebral disc by nerve growth factor-dependent neurons related to inflammatory pain. *Spine*. 2004;29:1077-1081.
159. Edgar MA. The nerve supply of the lumbar intervertebral disc. *J Bone Joint Surg Br*. 2007;89:1135-1139.
160. Jarvik JG, et al. Three-year incidence of low back pain in an initially asymptomatic cohort: clinical and imaging risk factors. *Spine*. 2005;30:1541-1548.
161. Carragee EJ, et al. Discographic, MRI and psychosocial determinants of low back pain disability and remission: a prospective study in subjects with benign persistent back pain. *Spine J*. 2005;5:24-35.
162. Carragee EJ, Paragioudakis SJ, Khurana S. 2000 Volvo Award winner in clinical studies: Lumbar high-intensity zone and discography in subjects without low back problems. *Spine*. 2000;25:2987-2992.
163. Carragee EJ, Alamin TF, Carragee JM. Low-pressure positive discography in subjects asymptomatic of significant low back pain illness. *Spine*. 2006;31:505-509.
164. Carragee EJ, et al. The rates of false-positive lumbar discography in select patients without low back symptoms. *Spine*. 2000;25:1373-1380.
165. O'Neill CW, et al. Disc stimulation and patterns of referred pain. *Spine*. 2002;27:2776-2781.

166. Deyo RA, Diehl AK. Lumbar spine films in primary care: current use and effects of selective ordering criteria. *J Gen Intern Med*. 1986;1:20-25.
167. Torgerson WR, Dotter WE. Comparative roentgenographic study of the asymptomatic and symptomatic lumbar spine. *J Bone Joint Surg Am*. 1976;58:850-853.
168. Liang M, Komaroff AL. Roentgenograms in primary care patients with acute low back pain: a cost-effectiveness analysis. *Arch Intern Med*. 1982;142:1108-1112.
169. Scavone JG, Latshaw RF, Weidner WA. Anteroposterior and lateral radiographs: an adequate lumbar spine examination. *AJR Am J Roentgenol*. 1981;136:715-717.
170. Institute for Clinical Systems Improvement. *Adult low back pain*; 2008. Institute for Clinical Systems Improvement. Bloomington, MN.
171. Aprill C, Bogduk N. High-intensity zone: a diagnostic sign of painful lumbar disc on magnetic resonance imaging. *Br J Radiol*. 1992;65:361-369.
172. Yu SW, et al. Tears of the anulus fibrosus: correlation between MR and pathologic findings in cadavers. *AJNR Am J Neuroradiol*. 1988;9:367-370.
173. Osti OL, Vernon-Roberts B, Moore R, Fraser RD. Annular tears and disc degeneration in the lumbar spine. A post-mortem study of 135 discs. *J Bone Joint Surg Br*. 1992;74:678-682.
174. Morgan S, Saifuddin A. MRI of the lumbar intervertebral disc. *Clin Radiol*. 1999;54:703-723.
175. Modic MT, et al. Degenerative disc disease: assessment of changes in vertebral body marrow with MR imaging. *Radiology*. 1988;166:193-199.
176. Boden SD, et al. Abnormal magnetic-resonance scans of the lumbar spine in asymptomatic subjects. A prospective investigation. *J Bone Joint Surg Am*. 1990;72:403-408.
177. Jensen MC, et al. Magnetic resonance imaging of the lumbar spine in people without back pain. *N Engl J Med*. 1994;331:69-73.
178. Stadnik TW, et al. Annular tears and disc herniation: prevalence and contrast enhancement on MR images in the absence of low back pain or sciatica. *Radiology*. 1998;206:49-55.
179. Borenstein DG, et al. The value of magnetic resonance imaging of the lumbar spine to predict low-back pain in asymptomatic subjects : a seven-year follow-up study. *J Bone Joint Surg Am*. 2001;83-A:1306-1311.
180. Jarvik JJ, et al. The Longitudinal Assessment of Imaging and Disability of the Back (LAIDBack) study: baseline data. *Spine*. 2001;26:1158-1166.
181. Jarvik JG, et al. Rapid magnetic resonance imaging vs radiographs for patients with low back pain: a randomized controlled trial. *JAMA*. 2003;289:2810-2818.
182. Carragee E, et al. Are first-time episodes of serious LBP associated with new MRI findings? *Spine J*. 2006;6:624-635.
183. Lappalainen AK, et al. The diagnostic value of contrast-enhanced magnetic resonance imaging in the detection of experimentally induced anular tears in sheep. *Spine*. 2002;27:2806-2810.
184. Yoshida H, et al. Diagnosis of symptomatic disc by magnetic resonance imaging: T2-weighted and gadolinium-DTPA-enhanced T1-weighted magnetic resonance imaging. *J Spinal Disord Tech*. 2002;15:193-198.
185. Aprill C, Bogduk N. High-intensity zone: a diagnostic sign of painful lumbar disc on magnetic resonance imaging. *Br J Radiol*. 1992;65:361-369.
186. Lim CH, et al. Discogenic lumbar pain: association with MRI imaging and CT discography. *Eur J Radiol*. 2005;54:431-437.
187. Lam KS, Carlin D, Mulholland RC. Lumbar disc high-intensity zone: the value and significance of provocative discography in the determination of the discogenic pain source. *Eur Spine J*. 2000;9:36-41.
188. Schellhas KP, et al. Lumbar disc high-intensity zone. Correlation of magnetic resonance imaging and discography. *Spine*. 1996;21:79-86.
189. Peng B, et al. The pathogenesis and clinical significance of a high-intensity zone (HIZ) of lumbar intervertebral disc on MR imaging in the patient with discogenic low back pain. *Eur Spine J*. 2006;15:583-587.
190. Lei D, et al. Painful disc lesion: can modern biplanar magnetic resonance imaging replace discography? *J Spinal Disord Tech*. 2008;21:430-435.
191. Saifuddin A, et al. The value of lumbar spine magnetic resonance imaging in the demonstration of anular tears. *Spine*. 1998;23:453-457.
192. Kang CH, et al. Can magnetic resonance imaging accurately predict concordant pain provocation during provocative disc injection? *Skeletal Radiol*. 2009;38:877-885.
193. Ricketson R, Simmons JW, Hauser BO. The prolapsed intervertebral disc. The high-intensity zone with discography correlation. *Spine*. 1996;21:2758-2762.
194. Schneiderman G, et al. Magnetic resonance imaging in the diagnosis of disc degeneration: correlation with discography. *Spine*. 1987;12:276-281.
195. Ito M, et al. Predictive signs of discogenic lumbar pain on magnetic resonance imaging with discography correlation. *Spine*. 1998;23:1252-1258.
196. Simmons JW, et al. Awake discography. A comparison study with magnetic resonance imaging. *Spine*. 1991;16(suppl 6): S216-S221.
197. Mitra D, Cassar-Pullicino VN, McCall IW. Longitudinal study of high intensity zones on MR of lumbar intervertebral discs. *Clin Radiol*. 2004;59:1002-1008.
198. Milette P, et al. Differentiating lumbar disc protrusions, disc bulges, and discs with normal contour but abnormal signal intensity: magnetic resonance imaging with discographic correlations. *Spine*. 1999;24:44-53.
199. Horton WC, Daftari TK. Which disc as visualized by magnetic resonance imaging is actually a source of pain? A correlation between magnetic resonance imaging and discography. *Spine*. 1992;17(suppl 6):S167-S171.
200. Toyone T, et al. Vertebral bone-marrow changes in degenerative lumbar disc disease. An MRI study of 74 patients with low back pain. *J Bone Joint Surg Br*. 1994;76:757-764.
201. Kuisma M, et al. Modic changes in endplates of lumbar vertebral bodies: prevalence and association with low back and sciatic pain among middle-aged male workers. *Spine*. 2007;32:1116-1122.
202. Albert HB, Manniche C. Modic changes following lumbar disc herniation. *Eur Spine J*. 2007;16:977-982.
203. Thompson KJ, et al. Modic changes on MR images as studied with provocative discography: clinical relevance—a retrospective study of 2457 discs. *Radiology*. 2009;250:849-855.
204. Mitra D, Cassar-Pullicino VN, McCall IW. Longitudinal study of vertebral type-1 end-plate changes on MR of the lumbar spine. *Eur Radiol*. 2004;14:1574-1581.
205. Marshman LA, et al. Reverse transformation of Modic type 2 changes to Modic type 1 changes during sustained chronic

low-back pain severity. Report of two cases and review of the literature. *J Neurosurg Spine*. 2007;6:152-155.
206. Kuisma M, et al. A three-year follow-up of lumbar spine endplate (Modic) changes. *Spine*. 2006;31:1714-1718.
207. Sandhu HS, et al. Association between findings of provocative discography and vertebral endplate signal changes as seen on MRI. *J Spinal Disord*. 2000;13:438-443.
208. Braithwaite I, et al. Vertebral end-plate (Modic) changes on lumbar spine MRI: correlation with pain reproduction at lumbar discography. *Eur Spine J*. 1998;7:363-368.
209. Kokkonen SM, et al. Endplate degeneration observed on magnetic resonance imaging of the lumbar spine: correlation with pain provocation and disc changes observed on computed tomography discography. *Spine*. 2002;27:2274-2278.
210. Schenk P, et al. Magnetic resonance imaging of the lumbar spine: findings in female subjects from administrative and nursing professions. *Spine*. 2006;31:2701-2706.
211. Kjaer P, et al. Magnetic resonance imaging and low back pain in adults: a diagnostic imaging study of 40-year-old men and women. *Spine*. 2005;30:1173-1180.
212. Kjaer P, Korsholm L, Bendix T, Sorensen JS, Leboeuf-Yde C. Modic changes and their associations with clinical findings. *Eur Spine J*. 2006;15:1312-1319.
213. Leboeuf-Yde C, et al. Self-reported hard physical work combined with heavy smoking or overweight may result in so-called Modic changes. *BMC Musculoskelet Disord*. 2008;9:5.
214. Jensen TS, et al. Vertebral endplate signal changes (Modic change): a systematic literature review of prevalence and association with non-specific low back pain. *Eur Spine J*. 2008;17:1407-1422.
215. Huang KY, et al. Factors affecting disability and physical function in degenerative lumbar spondylolisthesis of L4-5: evaluation with axially loaded MRI. *Eur Spine J*. 2009;18:1851-1857.
216. Hansson T, et al. The narrowing of the lumbar spinal canal during loaded MRI: the effects of the disc and ligamentum flavum. *Eur Spine J*. 2009;18:679-686.
217. Jayakumar P, et al. Dynamic degenerative lumbar spondylolisthesis: diagnosis with axial loaded magnetic resonance imaging. *Spine*. 2006;31:E298-E301.
218. Hiwatashi A, et al. Axial loading during MR imaging can influence treatment decision for symptomatic spinal stenosis. *AJNR Am J Neuroradiol*. 2004;25:170-174.
219. Danielson B, Willén J. Axially loaded magnetic resonance image of the lumbar spine in asymptomatic individuals. *Spine*. 2001;26:2601-2606.
220. Saifuddin A, McSweeney E, Lehovsky J. Development of lumbar high intensity zone on axial loaded magnetic resonance imaging. *Spine*. 2003;28:E449-E451.
221. Holt EP Jr. The question of lumbar discography. *J Bone Joint Surg Am*. 1968;50:720-726.
222. Carragee EJ, et al. Does discography cause accelerated progression of degeneration changes in the lumbar disc: a ten-year matched cohort study. *Spine*. 2009;34(21):2338-2345.
223. Chou R, et al. Interventional therapies, surgery, and interdisciplinary rehabilitation for low back pain: an evidence-based clinical practice guideline from the American Pain Society. *Spine*. 2009;34:1066-1077.
224. Aprill C. Diagnostic Disc Injection. In: Frymoyer J, ed. *The Adult Spine: Principles and Practice*. New York: Raven Press; 1991.
225. Guyer RD, Ohnmeiss DD. Lumbar discography. Position statement from the North American Spine Society Diagnostic and Therapeutic Committee. *Spine*. 1995;20:2048-2059.
226. Bernard TN Jr. Lumbar discography followed by computed tomography. Refining the diagnosis of low-back pain. *Spine*. 1990;15:690-707.
227. Massie W. A critical evaluation of discography: scientific exhibit. *J Bone Joint Surg Am*. 1967;49:1243-1244.
228. Walsh TR, et al. Lumbar discography in normal subjects. A controlled, prospective study. *J Bone Joint Surg Am*. 1990;72:1081-1088.
229. Derby R, et al. Comparison of discographic findings in asymptomatic subject discs and the negative discs of chronic LBP patients: can discography distinguish asymptomatic discs among morphologically abnormal discs? *Spine J*. 2005;5:389-394.
230. Buenaventura RM, et al. Systematic review of discography as a diagnostic test for spinal pain: an update. *Pain Physician*. 2007;10:147-164.
231. Wolfer LR, et al. Systematic review of lumbar provocation discography in asymptomatic subjects with a meta-analysis of false-positive rates. *Pain Physician*. 2008;11:513-538.
232. Colhoun E, et al. Provocation discography as a guide to planning operations on the spine. *J Bone Joint Surg Br*. 1988;70:267-271.
233. Madan S, et al. Does provocative discography screening of discogenic back pain improve surgical outcome? *J Spinal Disord Tech*. 2002;15:245-251.
234. Carragee EJ, et al. A gold standard evaluation of the "discogenic pain" diagnosis as determined by provocative discography. *Spine*. 2006;31:2115-2123.
235. Ohtori S, et al. Results of surgery for discogenic low back pain: a randomized study using discography versus discoblock for diagnosis. *Spine*. 2009;34:1345-1348.
236. Deyo RA. Conservative therapy for low back pain. Distinguishing useful from useless therapy. *JAMA*. 1983;250:1057-1062.
237. Moss IL, An HS, Shen FH, et al. The nonsurgical treatment of back pain. In: Shapiro IM, Risbud MV, eds. *The Intervertebral Disc*. New York: Springer-Verlag; 2014:247-259.
238. Deyo RA, Diehl AK, Rosenthal M. How many days of bed rest for acute low back pain? A randomized clinical trial. *N Engl J Med*. 1986;315:1064-1070.
239. Quinet RJ, Hadler NM. Diagnosis and treatment of backache. *Semin Arthritis Rheum*. 1979;8:261-287.
240. Rowe ML. Low back pain in industry. A position paper. *J Occup Med*. 1969;11:161-169.
241. Allen C, Glasziou P, Del Mar C. Bed rest: a potentially harmful treatment needing more careful evaluation. *Lancet*. 1999;354:1229-1233.
242. Hagen KB, et al. Bed rest for acute low-back pain and sciatica. *Cochrane Database Syst Rev*. 2004;(8):CD001254.
243. Verbunt JA, et al. A new episode of low back pain: who relies on bed rest? *Eur J Pain*. 2008;12:508-516.
244. Engers A, et al. Individual patient education for low back pain. *Cochrane Database Syst Rev*. 2008;(1):CD004057.
245. Liddle SD, Gracey JH, Baxter GD. Advice for the management of low back pain: a systematic review of randomised controlled trials. *Man Ther*. 2007;12:310-327.
246. Brox JI, et al. Evidence-informed management of chronic low back pain with back schools, brief education, and fear-avoidance training. *Spine J*. 2008;8:28-39.

247. Calmels P, et al. Effectiveness of a lumbar belt in subacute low back pain: an open, multicentric, and randomized clinical study. *Spine*. 2009;34:215-220.
248. Oleske DM, et al. Are back supports plus education more effective than education alone in promoting recovery from low back pain? Results from a randomized clinical trial. *Spine*. 2007;32:2050-2057.
249. van Duijvenbode IC, et al. Lumbar supports for prevention and treatment of low back pain. *Cochrane Database Syst Rev*. 2008;(2):CD001823.
250. Sahar T, et al. Insoles for prevention and treatment of back pain. *Cochrane Database Syst Rev*. 2007;(4):CD005275.
251. Brox JI, et al. Randomized clinical trial of lumbar instrumented fusion and cognitive intervention and exercises in patients with chronic low back pain and disc degeneration. *Spine*. 2003;28:1913-1921.
252. Keller A, et al. Trunk muscle strength, cross-sectional area, and density in patients with chronic low back pain randomized to lumbar fusion or cognitive intervention and exercises. *Spine*. 2004;29:3-8.
253. Fairbank J, Frost H, Wilson-MacDonald J, et al. Randomised controlled trial to compare surgical stabilisation of the lumbar spine with an intensive rehabilitation programme for patients with chronic low back pain: the MRC spine stabilisation trial. *BMJ*. 2005;330:1233.
254. Eleswarapu AS, et al. How effective is physical therapy for common low back pain diagnosis. *Spine*. 2016;41(16): 1325-1329.
255. Heymans MW, et al. The effectiveness of high-intensity versus low-intensity back schools in an occupational setting: a pragmatic randomized controlled trial. *Spine*. 2006;31:1075-1082.
256. Kääpä EH, et al. Multidisciplinary group rehabilitation versus individual physiotherapy for chronic nonspecific low back pain: a randomized trial. *Spine*. 2006;31:371-376.
257. Heymans MW, van Tulder MW, Esmail R, Bombardier C, Koes BW. Back schools for non-specific low-back pain. *Cochrane Database Syst Rev*. 2004;(4):CD000261.
258. van der Roer N, et al. Economic evaluation of an intensive group training protocol compared with usual care physiotherapy in patients with chronic low back pain. *Spine*. 2008;33:445-451.
259. van der Roer N, et al. Intensive group training protocol versus guideline physiotherapy for patients with chronic low back pain: a randomised controlled trial. *Eur Spine J*. 2008;17:1193-1200.
260. Poitras S, Brosseau L. Evidence-informed management of chronic low back pain with transcutaneous electrical nerve stimulation, interferential current, electrical muscle stimulation, ultrasound, and thermotherapy. *Spine J*. 2008;8:226-233.
261. Khadilkar A, et al. Transcutaneous electrical nerve stimulation (TENS) versus placebo for chronic low-back pain. *Cochrane Database Syst Rev*. 2008;(4):CD003008.
262. van Middlekoop M, Rubinstein SM, Kuipers T, et al. A systemic review on the effectiveness of physical and rehabilitation interventions for chronic non-specific low back pain. *Eur Spine J*. 2011;20(1):19-39.
263. Hoiriis KT, et al. A randomized clinical trial comparing chiropractic adjustments to muscle relaxants for subacute low back pain. *J Manipulative Physiol Ther*. 2004;27:388-398.
264. Haas M, Groupp E, Kraemer DF. Dose-response for chiropractic care of chronic low back pain. *Spine J*. 2004;4: 574-583.
265. McMorland G, Suter E. Chiropractic management of mechanical neck and low-back pain: a retrospective, outcome-based analysis. *J Manipulative Physiol Ther*. 2000;23:307-311.
266. Ernst E, Canter PH. A systematic review of systematic reviews of spinal manipulation. *J R Soc Med*. 2006;99:192-196.
267. Assendelft WJ, et al. Spinal manipulative therapy for low back pain. *Cochrane Database Syst Rev*. 2004;(1):CD000447.
268. Eisenberg DM, et al. Addition of choice of complementary therapies to usual care for acute low back pain: a randomized controlled trial. *Spine*. 2007;32:151-158.
269. Hurwitz EL, et al. A randomized trial of chiropractic and medical care for patients with low back pain: eighteen-month follow-up outcomes from the UCLA low back pain study. *Spine*. 2006;31(6):611-621.
270. Andersson GBJ, Lucente T, Davos AM, et al. A comparison of osteopathic spinal manipulation therapy and usual care for patients with subchronic low back pain. *N Engl J Med*. 1999;341:1126-1431.
271. Furlan AD, et al. Acupuncture and dry-needling for low back pain. *Cochrane Database Syst Rev*. 2005;(1):CD001351.
272. Ammendolia C, et al. Evidence-informed management of chronic low back pain with needle acupuncture. *Spine J*. 2008;8:160-172.
273. Yelland MJ, et al. Prolotherapy injections, saline injections, and exercises for chronic low-back pain: a randomized trial. *Spine*. 2004;29:9-16.
274. Dagenais S, et al. Evidence-informed management of chronic low back pain with prolotherapy. *Spine J*. 2008;8:203-212.
275. Rabago D, et al. A systematic review of prolotherapy for chronic musculoskeletal pain. *Clin J Sport Med*. 2005;15:376-380.
276. Roelofs PD, et al. Non-steroidal anti-inflammatory drugs for low back pain. *Cochrane Database Syst Rev*. 2008;(1):CD000396.
277. Deshpande A, et al. Opioids for chronic low-back pain. *Cochrane Database Syst Rev*. 2007;(3):CD004959.
278. Jamison RN, et al. Opioid therapy for chronic noncancer back pain. A randomized prospective study. *Spine*. 1998;23:2591-2600.
279. Martell BA, et al. Systematic review: opioid treatment for chronic back pain: prevalence, efficacy, and association with addiction. *Ann Intern Med*. 2007;146:116-127.
280. Schofferman J, Mazanec D. Evidence-informed management of chronic low back pain with opioid analgesics. *Spine J*. 2008;8:185-194.
281. Armstrong TA, Rohal GM. Potential danger from too much acetaminophen in opiate agonist combination products. *Am J Health Syst Pharm*. 1999;56:1774-1775.
282. *Physicians Desk Reference*. 63rd ed. Montvale, NJ: Thomson PDR; 2009.
283. Krenzelok EP. The FDA Acetaminophen Advisory Committee Meeting: what is the future of acetaminophen in the United States? The perspective of a committee member. *Clin Toxicol*. 2009;47:784-789.
284. Vamvanij V, et al. Surgical treatment of internal disc disruption: an outcome study of four fusion techniques. *J Spinal Disord*. 1998;11:375-382.
285. Chou R, Huffneu Lit. Medications for acute and chronic low back pain: a review of evidence for an American Pain Society clinical practice guideline. *Ann Intern Med*. 2007;147(7):492-504.

286. van Tulder MW, et al. Muscle relaxants for non-specific low back pain. *Cochrane Database Syst Rev*. 2003;(2):CD004252.
287. Alcoff J, et al. Controlled trial of imipramine for chronic low back pain. *J Fam Pract*. 1982;14:841-846.
288. Pheasant H, et al. Amitriptyline and chronic low-back pain. A randomized double-blind crossover study. *Spine*. 1983;8:552-557.
289. Ward NG. Tricyclic antidepressants for chronic low-back pain. Mechanisms of action and predictors of response. *Spine*. 1986;11:661-665.
290. Staiger TO, et al. Systematic review of antidepressants in the treatment of chronic low back pain. *Spine*. 2003;28: 250-255.
291. Chang V, Gonzalez P, Akuthota V. Evidence-informed management of chronic low back pain with adjunctive analgesics. *Spine J*. 2008;8:21-27.
292. Keller A, et al. Effect sizes of non-surgical treatments of non-specific low-back pain. *Eur Spine J*. 2007;16: 1776-1788.
293. Machado LA, et al. Analgesic effects of treatments for non-specific low back pain: a meta-analysis of placebo-controlled randomized trials. *Rheumatology*. 2009;48:520-527.
294. Delaney TJ, et al. Epidural steroid effects on nerves and meninges. *Anesth Analg*. 1980;59:610-614.
295. Fairbank JC, et al. Apophyseal injection of local anesthetic as a diagnostic aid in primary low-back pain syndromes. *Spine*. 1981;6:598-605.
296. Manchikanti L, et al. Preliminary results of a randomized, equivalence trial of fluoroscopic caudal epidural injections in managing chronic low back pain. Part 1: discogenic pain without disc herniation or radiculitis. *Pain Physician*. 2008;11:785-800.
297. Manchikanti L, et al. Effectiveness of caudal epidural injections in discogram positive and negative chronic low back pain. *Pain Physician*. 2002;5:18-29.
298. Manchikanti L, et al. Caudal epidural injections with sarapin or steroids in chronic low back pain. *Pain Physician*. 2001;4:322-335.
299. Buttermann GR. The effect of spinal steroid injections for degenerative disc disease. *Spine J*. 2004;4:495-505.
300. Rosenberg SK, et al. Effectiveness of transforaminal epidural steroid injections in low back pain: a one year experience. *Pain Physician*. 2002;5:266-270.
301. DePalma MJ, Slipman CW. Evidence-informed management of chronic low back pain with epidural steroid injections. *Spine J*. 2008;8:45-55.
302. Staal JB, et al. Injection therapy for subacute and chronic low-back pain. *Cochrane Database Syst Rev*. 2008;(3):CD001824.
303. Chou R, et al. Nonsurgical interventional therapies for low back pain: a review of the evidence for an American Pain Society clinical practice guideline. *Spine*. 2009;34:1078-1093.
304. Feffer HL. Therapeutic intradiscal hydrocortisone. A long-term study. *Clin Orthop Relat Res*. 1969;67:100-104.
305. Wilkinson HA, Schuman N. Intradiscal corticosteroids in the treatment of lumbar and cervical disc problems. *Spine*. 1980;5:385-389.
306. Fayad F, et al. Relation of inflammatory Modic changes to intradiscal steroid injection outcome in chronic low back pain. *Eur Spine J*. 2007;16:925-931.
307. Simmons JW, et al. Intradiscal steroids. A prospective double-blind clinical trial. *Spine*. 1992;17(suppl 6):S172-S175.
308. Khot A, et al. The use of intradiscal steroid therapy for lumbar spinal discogenic pain: a randomized controlled trial. *Spine*. 2004;29:833-836.
309. Klein RG, et al. Biochemical injection treatment for discogenic low back pain: a pilot study. *Spine J*. 2003;3:220-226.
310. Miller MR, Mathews RS, Reeves KD. Treatment of painful advanced internal lumbar disc derangement with intradiscal injection of hypertonic dextrose. *Pain Physician*. 2006;9:115-121.
311. Peng B, et al. Intradiscal methylene blue injection for the treatment of chronic discogenic low back pain. *Eur Spine J*. 2007;16:33-38.
312. Gallucci M, et al. Sciatica: treatment with intradiscal and intraforaminal injections of steroid and oxygen-ozone versus steroid only. *Radiology*. 2007;242:907-913.
313. Muto M, et al. Low back pain and sciatica: treatment with intradiscal-intraforaminal O(2)-O (3) injection. Our experience. *Radiol Med*. 2008;113:695-706.
314. Freeman BJ, et al. Does intradiscal electrothermal therapy denervate and repair experimentally induced posterolateral annular tears in an animal model? *Spine*. 2003;28:2602-2608.
315. Shah RV, et al. Intradiscal electrothermal therapy: a preliminary histologic study. *Arch Phys Med Rehabil*. 2001;82:1230-1237.
316. Kleinstueck FS, et al. Acute biomechanical and histological effects of intradiscal electrothermal therapy on human lumbar discs. *Spine*. 2001;26:2198-2207.
317. Strohbeln JW. Temperature distributions from interstitial RF electrode hyperthermia systems: theoretical predictions. *Int J Radiat Oncol Biol Phys*. 1983;9:1655-1667.
318. Troussier B, et al. Percutaneous intradiscal radio-frequency thermocoagulation. A cadaveric study. *Spine*. 1995;20:1713-1718.
319. Houpt JC, Conner ES, McFarland EW. Experimental study of temperature distributions and thermal transport during radiofrequency current therapy of the intervertebral disc. *Spine*. 1996;21:1808-1812.
320. Ashley J, Gharpuray V, Saal J. Temperature distribution in the intervertebral disc: a comparison of intranuclear radiofrequency needle to a novel heating catheter. Proceedings of the 1999 Bioengineering Conference. 1999;42:77.
321. Karasek M, Bogduk N. Twelve-month follow-up of a controlled trial of intradiscal thermal anuloplasty for back pain due to internal disc disruption. *Spine*. 2000; 25:2601-2607.
322. Hsia AW, Isaac K, Katz JS. Cauda equina syndrome from intradiscal electrothermal therapy. *Neurology*. 2000;55:320.
323. Bogduk N, Karasek M. Two-year follow-up of a controlled trial of intradiscal electrothermal anuloplasty for chronic low back pain resulting from internal disc disruption. *Spine J*. 2002;2:343-350.
324. Saal JA, Saal JS. Intradiscal electrothermal treatment for chronic discogenic low back pain: prospective outcome study with a minimum 2-year follow-up. *Spine*. 2002;27:966-973.
325. Freeman BJ, et al. A randomized, double-blind, controlled trial: intradiscal electrothermal therapy versus placebo for the treatment of chronic discogenic low back pain. *Spine*. 2005;30:2369-2377.
326. Pauza KJ, et al. A randomized, placebo-controlled trial of intradiscal electrothermal therapy for the treatment of discogenic low back pain. *Spine J*. 2004;4:27-35.

327. Barendse GA, et al. Randomized controlled trial of percutaneous intradiscal radiofrequency thermocoagulation for chronic discogenic back pain: lack of effect from a 90-second 70 C lesion. *Spine*. 2001;26:287-292.
328. Andersson GB, Mekhail NA, Block JE. Treatment of intractable discogenic low back pain. A systematic review of spinal fusion and intradiscal electrothermal therapy (IDET). *Pain Physician*. 2006;9:237-248.
329. Derby R, et al. Evidence-informed management of chronic low back pain with intradiscal electrothermal therapy. *Spine J*. 2008;8:80-95.
330. Helm S, et al. Systematic review of the effectiveness of thermal annular procedures in treating discogenic low back pain. *Pain Physician*. 2009;12:207-232.
331. Freeman BJ. IDET: a critical appraisal of the evidence. *Eur Spine J*. 2006;15(suppl 3):S448-S457.
332. Chou R, et al. Surgery for low back pain: a review of the evidence for an American Pain Society Clinical Practice Guideline. *Spine*. 2009;34:1094-1109.
333. Deyo RA, Nachemson A, Mirza SK. Spinal-fusion surgery—the case for restraint. *N Engl J Med*. 2004;350:722-726.
334. Fritzell P, et al. Lumbar fusion versus nonsurgical treatment for chronic low back pain: a multicenter randomized controlled trial from the Swedish Lumbar Spine Study Group. *Spine*. 2001;26:2521-2532.
335. Rivero-Arias O, et al. Surgical stabilisation of the spine compared with a programme of intensive rehabilitation for the management of patients with chronic low back pain: cost utility analysis based on a randomised controlled trial. *BMJ*. 2005;330:1239.
336. Ibrahim T, Tleyjeh IM, Gabbar O. Surgical versus non-surgical treatment of chronic low back pain: a meta-analysis of randomised trials. *Int Orthop*. 2008;32:107-113.
337. Mirza SK, Deyo RA. Systematic review of randomized trials comparing lumbar fusion surgery to nonoperative care for treatment of chronic back pain. *Spine*. 2007;32:816-823.
338. Phillips FM, Slosar PJ, Youssef JA, Andersson G. Papatheofaris F. Lumbar spine fusion for chronic low back pain due to degenerative disc disease. *Spine*. 2013;38(7):409-422.
339. Anderson PA, Andersson GBJ, Arnold PM, et al. Terminology. *Spine*. 2012;37(22s):58-59.
340. Lund T, Oxland TR. Adjacent level disc disease – is it really fusion disease? *Orthop Clinic North Am*. 2011;42: 529-541.
341. Bae WC, Masuda K. Emerging technologies for molecular therapy for intervertebral disc degeneration. *Ortrhop Clin North Am*. 2011;42:585-601.
342. Woods BL, Vo N, Sowa G, Kang JD. Gene therapy for intervertebral disc degeneration. *Orthop Clin North Am*. 2011;42:563-574.
343. Sakai D. Stem cell regeneration of the intervertebral disc. *Orthop Clin North Am*. 2011;42:555-562.
344. Leung VYL, Tam V, Chan D, Chen BP, Cheung KMC. Tissue engineering for intervertebral disc degeneration. *Orthop Clin North Am*. 2011;42:575-583.